



Top 48 CAT Venn Diagrams Questions With Video Solutions

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Questions

Instructions

For the following questions answer them individually

Question 1

Choose the set in which the statements are most logically related.

- A. No fishes breathe through lungs.
- B. All fishes have scales.
- C. Some fishes breed up stream.
- D. All whales breathe through lungs.
- E. No whales are fishes.
- F. All whales are mammals.

- A ABC
- B BCD
- C ADE
- D DEF

 [VIEW SOLUTION](#)

Question 2

Choose the set in which the statements are most logically related.

- A. All roses are fragrant.
- B. All roses are majestic.
- C. All roses are plants.
- D. All roses need air.
- E. All plants need air.
- F. All plants need water.

- A CED
- B ACB
- C BDC
- D CFE

 [VIEW SOLUTION](#)

Question 3

Choose the set in which the statements are most logically related.

- A. No lunatics are fit to serve on a jury.
- B. Everyone who is sane can do logic.
- C. None of your sons can do logic.
- D. Some who can do logic are fit to serve on a jury.
- E. All who can do logic are fit to serve on a jury.
- F. Everyone who is sane is fit to serve on a jury.

- A BDE

- B BEF
- C BDF
- D ADE

[VIEW SOLUTION](#)



Question 4

Choose the set in which the statements are most logically related.

- A. All candid men are persons who acknowledge merit in a rival.
- B. Some learned men are very candid.
- C. Some learned men are not persons who acknowledge merit in a rival.
- D. Some learned men are persons who are very candid.
- E. Some learned men are not candid.
- F. Some persons who recognize merit in a rival are learned.

- A ABE
- B ACF
- C ADE
- D BAF

[VIEW SOLUTION](#)

Question 5

Choose the set in which the statements are most logically related.

- A. All those who achieve great ends are happy.
- B. All young people are happy.
- C. All young people achieve great ends.
- D. No young people achieve great ends.
- E. No young people are happy.
- F. Some young people are happy.

- A ADE
- B ABF
- C ACB
- D ADF

[VIEW SOLUTION](#)

Question 6

Choose the set in which the statements are most logically related.

- A. Some beliefs are uncertain.
- B. Nothing uncertain is worth dying for.

- C. Some belief is worth dying for.
- D. All beliefs are uncertain.
- E. Some beliefs are certain.
- F. No belief is worth dying for.

- A ABF
- B BCD
- C BEF
- D BDF

[VIEW SOLUTION](#)



Question 7

Choose the set in which the statements are most logically related.

- A. Some of my closest friends disapprove of me.
- B. Some of my closest friends are aardvarks.
- C. All of my closest friends disapprove of me.
- D. All who disapprove of me are aardvarks.
- E. Some who disapprove of me are aardvarks.
- F. Some of my closest friends are no aardvarks.

- A BCD
- B ABD
- C BCE
- D ABE

[VIEW SOLUTION](#)

Question 8

Choose the set in which the statements are most logically related.

- A. First-year students of this college like to enter for the prize.
- B. All students of this college rank as University students.
- C. First-year students of this college are entitled to enter for the prize.
- D. Some who rank as University students are First-year students.
- E. All University students are eligible to enter for the prize.
- F. All those who like to are entitled to enter for the prize.

- A AEF
- B ABC
- C BEC
- D CDF

 [VIEW SOLUTION](#)

Question 9

Choose the set in which the statements are most logically related.

- A. Some mammals are carnivores.
- B. All whales are mammals.
- C. All whales are aquatic animals.
- D. All whales are carnivores.
- E. Some aquatic animals are mammals.
- F. Some mammals are whales.

A ADF

B ABC

C AEF

D BCE

 [VIEW SOLUTION](#)



CAT Previous Papers

with detailed video solutions and analysis



Question 10

Choose the set in which the statements are most logically related.

- A. All men are men of scientific ability.
- B. Some women are women of scientific ability.
- C. Some men are men of artistic genius.
- D. Some men and women are of scientific ability.
- E. All men of artistic genius are men of scientific ability.
- F. Some women of artistic genius are women of scientific ability.

A ACD

B ACE

C DEF

D ABC

 [VIEW SOLUTION](#)

Instructions

Applicants for the doctoral programmes of Ambi Institute of Engineering (AIE) and Bambi Institute of Engineering (BIE) have to appear for a Common Entrance Test (CET). The test has three sections: Physics (P), Chemistry (C), and Maths (M). Among those appearing for CET, those at or above the 80th percentile in at least two sections, and at or above the 90th percentile overall, are selected for Advanced Entrance Test (AET) conducted by AIE. AET is used by AIE for final selection.

For the 200 candidates who are at or above the 90th percentile overall based on CET, the following are known about their performance in CET:

1. No one is below the 80th percentile in all 3 sections.

2. 150 are at or above the 80th percentile in exactly two sections.
3. The number of candidates at or above the 80th percentile only in P is the same as the number of candidates at or above the 80th percentile only in C. The same is the number of candidates at or above the 80th percentile only in M.
4. Number of candidates below 80th percentile in P: Number of candidates below 80th percentile in C: Number of candidates below 80th percentile in M = 4:2:1.

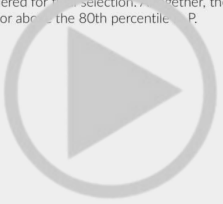
BIE uses a different process for selection. If any candidate is appearing in the AET by AIE, BIE considers their AET score for final selection provided the candidate is at or above the 80th percentile in P. Any other candidate at or above the 80th percentile in P in CET, but who is not eligible for the AET, is required to appear in a separate test to be conducted by BIE for being considered for final selection. Altogether, there are 400 candidates this year who are at or above the 80th percentile in P.

Question 11

If the number of candidates who are at or above the 90th percentile overall and also at or above the 80th percentile in all three sections in CET is actually a multiple of 5, what is the number of candidates who are at or above the 90th percentile overall and at or above the 80th percentile in both P and M in CET?



BIE uses a different process for selection. If any candidate is appearing in the AET by AIE, BIE considers their AET score for final selection provided the candidate is at or above the 80th percentile in P. Any other candidate at or above the 80th percentile in P in CET, but who is not eligible for the AET, is required to appear in a separate test to be conducted by BIE for being considered for final selection. Altogether, there are 400 candidates this year who are at or above the 80th percentile in P.





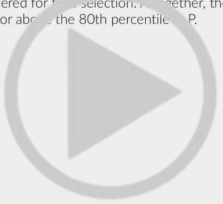
 VIDEO SOLUTION

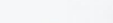
Question 12

If the number of candidates who are at or above the 90th percentile overall and also at or above the 80th percentile in all three sections in CET is actually a multiple of 5, then how many candidates were shortlisted for the AET for AIE?



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CAT Syllabus PDF



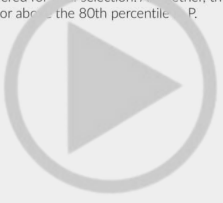
Question 13

If the number of candidates who are at or above the 90th percentile overall and also are at or above the 80th percentile in P in CET, is more than 100, how many candidates had to sit for the separate test for BIE?

- A 299
- B 310
- C 321
- D 330



BIE uses a different process for selection. If any candidate is appearing in the AET by AIE, BIE considers their AET score for final selection provided the candidate is at or above the 80th percentile in P. Any other candidate at or above the 80th percentile in P in CET, but who is not eligible for the AET, is required to appear in a separate test to be conducted by BIE for being considered for final selection. Together, there are 400 candidates this year who are at or above the 80th percentile in P.





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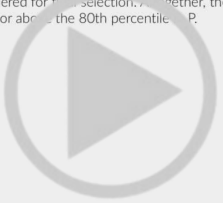
Question 14

What best can be concluded about the number of candidates sitting for the separate test for BIE who were at or above the 90th percentile overall in CET?

- A 3 or 10
- B 10
- C 5
- D 7 or 10



BIE uses a different process for selection. If any candidate is appearing in the AET by AIE, BIE considers their AET score for final selection provided the candidate is at or above the 80th percentile in P. Any other candidate at or above the 80th percentile in P in CET, but who is not eligible for the AET, is required to appear in a separate test to be conducted by BIE for being considered for final selection. Together, there are 400 candidates this year who are at or above the 80th percentile in P.





 VIDEO SOLUTION

Instructions

1. The numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1.
2. The number of satellites serving all three of B, C, and S is 100.
3. The number of satellites exclusively serving C is the same as the number of satellites exclusively serving S. This number is 30% of the number of satellites exclusively serving B.
4. The number of satellites serving O is the same as the number of satellites serving both C and S but not B.

If at least 100 of the 1600 satellites were serving O, what can be said about the number of satellites serving S?

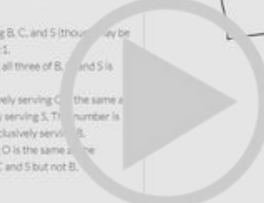
- A** At most 475
- B** Exactly 475
- C** No conclusion is possible based on the given information
- D** At least 475

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1400 satellites were sent up by a country for several purposes. The purposes are classified as broadcasting (B), communication (C), surveillance (S), and others (O). A satellite can serve multiple purposes; however a satellite serving either B or C, or S does not serve O. The following facts are known about the satellites:



1. The numbers of satellites serving B, C, and S (though they are not exclusively) are in the ratio 2:1:1.
2. The number of satellites serving all three of B, C, and S is 100.
3. The number of satellites exclusively serving C is the same as the number of satellites exclusively serving S. The number is 30% of the number of satellites exclusively serving B.
4. The number of satellites serving O is the same as the number of satellites serving both C and S but not B.





CAT Formula PDF


If the number of satellites serving at least two among B, C, and S is 1200, which of the following MUST be FALSE?

- A** The number of satellites serving B exclusively is exactly 250
- B** The number of satellites serving B is more than 1000
- C** The number of satellites serving C cannot be uniquely determined
- D** All 1600 satellites serve B or C or S

1. The numbers of satellites serving B, C, and S (though they may not exclusively) are in the ratio 2:1:1.
2. The number of satellites serving all three of B, C, and S is 100.
3. The number of satellites exclusively serving C is the same as the number of satellites exclusively serving S. The number is 30% of the number of satellites exclusively serving B.
4. The number of satellites serving O is the same as the number of satellites serving both C and S but not B.



Question 17

- | | |
|----------|-----|
| A | 250 |
| B | 100 |
| C | 500 |
| D | 200 |

A Venn diagram with three overlapping circles labeled B, C, and S. The circles B and C are at the top, overlapping each other, and both overlap with circle S below them. The entire diagram is enclosed in a rectangle labeled G in the bottom right corner.

1. The numbers of satellites serving B, C, and S (though not exclusively) are in the ratio 2:1:1.
2. The number of satellites serving all three of B, C, and S is 100.
3. The number of satellites exclusively serving C is the same as the number of satellites exclusively serving S. The number is 30% of the number of satellites exclusively serving B.
4. The number of satellites serving O is the same as the number of satellites serving both C and S but not B.



Question 18

about the number
st 100

- A** Must be at least 100
- B** Cannot be more than 800
- C** Must be between 450 and 725
- D** Must be between 400 and 800

1400 satellites were sent up by a country for several purposes. The purposes are classified as broadcasting (B), communication (C), surveillance (S), and others (O). A satellite can serve multiple purposes; however, a satellite serving either B or C or S does not serve O. The following facts are known about the satellites:

1. The numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1.
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4. The number of satellites serving O is the same as the number of satellites serving both C and S but not B.



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VIDEO SOLUTION

IIM Call Predictor
Accurate analysis of your profile




Instructions

Fun Sports (FS) provides training in three sports - Gilli-danda (G), Kho-Kho (K), and Ludo (L). Currently it has an enrollment of 39 students each of whom is enrolled in at least one of the three sports. The following details are known:

1. The number of students enrolled only in L is double the number of students enrolled in all the three sports.
2. There are a total of 17 students enrolled in G.
3. The number of students enrolled only in G is one less than the number of students enrolled only in L.
4. The number of students enrolled only in K is equal to the number of students who are enrolled in both K and L.
5. The maximum student enrollment is in L.
6. Ten students enrolled in G are also enrolled in at least one more sport.

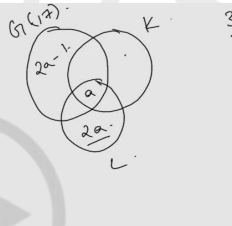
Question 19

Due to academic pressure, students who were enrolled in all three sports were asked to withdraw from one of the three sports. After the withdrawal, the number of students enrolled in G was six less than the number of students enrolled in L, while the number of students enrolled in K went down by one. After the withdrawal, how many students were enrolled in both G and K?

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Fun Sports (FS) provides training in three sports - Gilli-danda (G), Kho-Kho (K), and Ludo (L). Currently it has an enrollment of 39 students each of whom is enrolled in at least one of the three sports. The following details are known:

1. The number of students enrolled only in L is double the number of students enrolled in all the three sports.
2. There are a total of 17 students enrolled in G.
3. The number of students enrolled only in G is one less than the number of students enrolled only in L.
4. The number of students enrolled only in K is equal to the number of students who are enrolled in both K and L.
5. The maximum student enrollment is in L.
6. Ten students enrolled in G are also enrolled in at least one more sport.



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VIDEO SOLUTION

Question 20

Due to academic pressure, students who were enrolled in all three sports were asked to withdraw from one of the three sports. After the withdrawal, the number of students enrolled in G was six less than the number of students enrolled in L, while the number of students enrolled in K went down by one. After the withdrawal, how many students were enrolled in both G and L?

- A 6
- B 5
- C 7
- D 8

cracku

Fun Sports (FS) provides training in three sports - Gilli-danda (G), Kho-Kho (K), and Ludo (L). Currently it has an enrollment of 39 students each of whom is enrolled in at least one of the three sports. The following details are known:

1. The number of students enrolled only in L is double the number of students enrolled in all the three sports.
2. There are a total of 17 students enrolled in G.
3. The number of students enrolled only in G is one less than the number of students enrolled only in L.
4. The number of students enrolled only in K is equal to the number of students who are enrolled in both K and L.
5. The maximum student enrollment is in L.
6. Ten students enrolled in G are also enrolled in at least one more sport.

39

VIDEO SOLUTION

Question 21

If the numbers of students enrolled in K and L are in the ratio 19:22, then what is the number of students enrolled in L?

- A 18
- B 17
- C 22
- D 19

cracku

Fun Sports (FS) provides training in three sports - Gilli-danda (G), Kho-Kho (K), and Ludo (L). Currently it has an enrollment of 39 students each of whom is enrolled in at least one of the three sports. The following details are known:

1. The number of students enrolled only in L is double the number of students enrolled in all the three sports.
2. There are a total of 17 students enrolled in G.
3. The number of students enrolled only in G is one less than the number of students enrolled only in L.
4. The number of students enrolled only in K is equal to the number of students who are enrolled in both K and L.
5. The maximum student enrollment is in L.
6. Ten students enrolled in G are also enrolled in at least one more sport.

39

VIDEO SOLUTION

CAT Percentile Predictor



Question 22

What is the minimum number of students enrolled in both G and L but not in K?

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Fun Sports (FS) provides training in three sports - Gilli-danda (G), Kho-Kho (K), and Ludo (L). Currently it has an enrollment of 39 students each of whom is enrolled in at least one of the three sports. The following details are known:

1. The number of students enrolled only in L is double the number of students enrolled in all the three sports.
2. There are a total of 17 students enrolled in G.
3. The number of students enrolled only in G is one less than the number of students enrolled only in L.
4. The number of students enrolled only in K is equal to the number of students who are enrolled in both K and L.
5. The maximum student enrollment is in L.
6. Ten students enrolled in G are also enrolled in at least one more sport.

Handwritten Venn diagram showing three sets: G (Gilli-danda), K (Kho-Kho), and L (Ludo). The regions are labeled with variables: $2a-1$ for G only, a for K only, $2a$ for L only, a for G and K, a for G and L, a for K and L, and a for all three. The total number of students is 39.

VIDEO SOLUTION

Instructions

There are 15 girls and some boys among the graduating students in a class. They are planning a get-together, which can be either a 1-day event, or a 2-day event, or a 3-day event. There are 6 singers in the class, 4 of them are boys. There are 10 dancers in the class, 4 of them are girls. No dancer in the class is a singer.

Some students are not interested in attending the get-together. Those students who are interested in attending a 3-day event are also interested in attending a 2-day event; those who are interested in attending a 2-day event are also interested in attending a 1-day event.

The following facts are also known:

1. All the girls and 80% of the boys are interested in attending a 1-day event. 60% of the boys are interested in attending a 2-day event.
2. Some of the girls are interested in attending a 1-day event, but not a 2-day event; some of the other girls are interested in attending both.
3. 70% of the boys who are interested in attending a 2-day event are neither singers nor dancers. 60% of the girls who are interested in attending a 2-day event are neither singers nor dancers.
4. No girl is interested in attending a 3-day event. All male singers and 2 of the dancers are interested in attending a 3-day event.
5. The number of singers interested in attending a 2-day event is one more than the number of dancers interested in attending a 2-day event.

Question 23

What BEST can be concluded about the number of male dancers who are interested in attending a 1-day event?

- A 5 or 6
- B 6
- C 5
- D 4 or 6

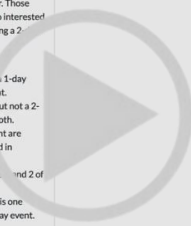
cracku

There are 15 girls and some boys among the graduating students in a class. They are planning a get-together, which can be either a 1-day event, or a 2-day event, or a 3-day event. There are 6 singers in the class, 4 of them are boys. There are 10 dancers in the class, 4 of them are girls. No dancer in the class is a singer.

Some students are not interested in attending the get-together. Those students who are interested in attending a 3-day event are also interested in attending a 2-day event; those who are interested in attending a 2-day event are also interested in attending a 1-day event.

The following facts are also known:

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3. 70% of the boys who are interested in attending a 2-day event are neither singers nor dancers. 60% of the girls who are interested in attending a 2-day event are neither singers nor dancers.
4. No girl is interested in attending a 3-day event. All male singers and 2 of the dancers are interested in attending a 3-day event.
5. The number of singers interested in attending a 2-day event is one more than the number of dancers interested in attending a 2-day event.




 VIDEO SOLUTION

Question 24

How many boys are there in the class?

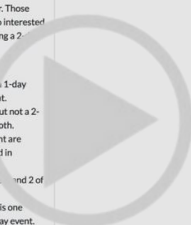
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There are 15 girls and some boys among the graduating students in a class. They are planning a get-together, which can be either a 1-day event, or a 2-day event, or a 3-day event. There are 6 singers in the class, 4 of them are boys. There are 10 dancers in the class, 4 of them are girls. No dancer in the class is a singer.

Some students are not interested in attending the get-together. Those students who are interested in attending a 3-day event are also interested in attending a 2-day event; those who are interested in attending a 2-day event are also interested in attending a 1-day event.

The following facts are also known:

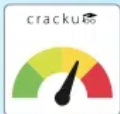
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2. Some of the girls are interested in attending a 1-day event but not a 2-day event; some of the other girls are interested in attending both.
3. 70% of the boys who are interested in attending a 2-day event are neither singers nor dancers. 60% of the girls who are interested in attending a 2-day event are neither singers nor dancers.
4. No girl is interested in attending a 3-day event. All male singers and 2 of the dancers are interested in attending a 3-day event.
5. The number of singers interested in attending a 2-day event is one more than the number of dancers interested in attending a 2-day event.




 VIDEO SOLUTION



CAT Score Calculator




Question 25

Which of the following can be determined from the given information?

- I. The number of boys who are interested in attending a 1-day event and are neither dancers nor singers.
- II. The number of female dancers who are interested in attending a 1-day event.

- A Only I
- B Neither I nor II
- C Only II
- D Both I and II

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There are 15 girls and some boys among the graduating students in a class. They are planning a get-together, which can be either a 1-day event, or a 2-day event, or a 3-day event. There are 6 singers in the class, 4 of them are boys. There are 10 dancers in the class, 4 of them are girls. No dancer in the class is a singer.

Some students are not interested in attending the get-together. Those students who are interested in attending a 3-day event are also interested in attending a 2-day event; those who are interested in attending a 2-day event are also interested in attending a 1-day event.

The following facts are also known:

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2. Some of the girls are interested in attending a 1-day event but not a 2-day event; some of the other girls are interested in attending both.
3. 70% of the boys who are interested in attending a 2-day event are neither singers nor dancers. 60% of the girls who are interested in attending a 2-day event are neither singers nor dancers.
4. No girl is interested in attending a 3-day event. All male singers and 2 of the dancers are interested in attending a 3-day event.
5. The number of singers interested in attending a 2-day event is one more than the number of dancers interested in attending a 2-day event.



VIDEO SOLUTION

Question 26

How many female dancers are interested in attending a 2-day event?

- A 2
- B 1
- C 0
- D Cannot be determined

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There are 15 girls and some boys among the graduating students in a class. They are planning a get-together, which can be either a 1-day event, or a 2-day event, or a 3-day event. There are 6 singers in the class, 4 of them are boys. There are 10 dancers in the class, 4 of them are girls. No dancer in the class is a singer.

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4. No girl is interested in attending a 3-day event. All male singers and 2 of the dancers are interested in attending a 3-day event.
5. The number of singers interested in attending a 2-day event is one more than the number of dancers interested in attending a 2-day event.



VIDEO SOLUTION

Question 27

What fraction of the class are interested in attending a 2-day event?

- A $\frac{7}{10}$
- B $\frac{7}{13}$
- C $\frac{9}{13}$
- D $\frac{2}{3}$

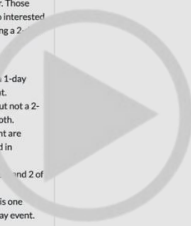
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1. All the girls and 80% of the boys are interested in attending a 1-day event. 60% of the boys are interested in attending a 2-day event.
2. Some of the girls are interested in attending a 1-day event but not a 2-day event; some of the other girls are interested in attending both.
3. 70% of the boys who are interested in attending a 2-day event are neither singers nor dancers. 60% of the girls who are interested in attending a 2-day event are neither singers nor dancers.
4. No girl is interested in attending a 3-day event. All male singers and 2 of the dancers are interested in attending a 3-day event.
5. The number of singers interested in attending a 2-day event is one more than the number of dancers interested in attending a 2-day event.




VIDEO SOLUTION

cracku

CAT Crash Course

Live Classes by IIMA alumni



Instructions

Students in a college are discussing two proposals --

A: a proposal by the authorities to introduce dress code on campus, and

B: a proposal by the students to allow multinational food franchises to set up outlets on college campus.

A student does not necessarily support either of the two proposals.

In an upcoming election for student union president, there are two candidates in fray:

Sunita and Ragini. Every student prefers one of the two candidates.

A survey was conducted among the students by picking a sample of 500 students. The following information was noted from this survey.

1. 250 students supported proposal A and 250 students supported proposal B.
2. Among the 200 students who preferred Sunita as student union president, 80% supported proposal A.
3. Among those who preferred Ragini, 30% supported proposal A.
4. 20% of those who supported proposal B preferred Sunita.
5. 40% of those who did not support proposal B preferred Ragini.
6. Every student who preferred Sunita and supported proposal B also supported proposal A.
7. Among those who preferred Ragini, 20% did not support any of the proposals.

Question 28

How many of the students surveyed supported proposal B, did not support proposal A and preferred Ragini as student union president?

- A 150
- B 210
- C 200
- D 40

cracku

Students in a college are discussing two proposals –
A: a proposal by the authorities to introduce dress code on campus,
 and
B: a proposal by the students to allow multinational food franchises to set up outlets on college campus.

A student does not necessarily support either of the two proposals.

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3. Among those who preferred Ragini, 30% supported proposal A.
4. 20% of those who supported proposal B preferred Sunita.
5. 40% of those who did not support proposal B preferred Ragini.
6. Every student who preferred Sunita and supported proposal B also supported proposal A.
7. Among those who preferred Ragini, 20% did not support any of the proposals.

$A = 250$
 $S = 200$
 $R = 300$
 $T = 500$



[VIDEO SOLUTION](#)

Question 29

What percentage of the students surveyed who supported both proposals A and B preferred Sunita as student union president?

- A 40
- B 25
- C 20
- D 50

cracku

Students in a college are discussing two proposals –
A: a proposal by the authorities to introduce dress code on campus,
 and
B: a proposal by the students to allow multinational food franchises to set up outlets on college campus.

A student does not necessarily support either of the two proposals.

In an upcoming election for student union president, there are two candidates in fray:
Sunita and Ragini. Every student prefers one of the two candidates.

A survey was conducted among the students by picking a sample of 500 students. The following information was noted from this survey:

1. 250 students supported proposal A and 250 students supported proposal B.
2. Among the 200 students who preferred Sunita as student union president, 80% supported proposal A.
3. Among those who preferred Ragini, 30% supported proposal A.
4. 20% of those who supported proposal B preferred Sunita.
5. 40% of those who did not support proposal B preferred Ragini.
6. Every student who preferred Sunita and supported proposal B also supported proposal A.
7. Among those who preferred Ragini, 20% did not support any of the proposals.

$A = 250$
 $S = 200$
 $R = 300$
 $T = 500$



[VIDEO SOLUTION](#)

Question 30

What percentage of the students surveyed who did not support proposal A preferred Ragini as student union president?

cracku

Students in a college are discussing two proposals –
A: a proposal by the authorities to introduce dress code on campus,
 and
B: a proposal by the students to allow multinational food franchises to set up outlets on college campus.

A student does not necessarily support either of the two proposals.

In an upcoming election for student union president, there are two candidates in fray:
Sunita and Ragini. Every student prefers one of the two candidates.

A survey was conducted among the students by picking a sample of 500 students. The following information was noted from this survey:

1. 250 students supported proposal A and 250 students supported proposal B.
2. Among the 200 students who preferred Sunita as student union president, 80% supported proposal A.
3. Among those who preferred Ragini, 30% supported proposal A.
4. 20% of those who supported proposal B preferred Sunita.
5. 40% of those who did not support proposal B preferred Sunita.
6. Every student who preferred Sunita and supported proposal B also supported proposal A.
7. Among those who preferred Ragini, 20% did not support any of the proposals.

$A = 250$
 $S = 200$
 $R = 300$
 $T = 500$



VIDEO SOLUTION

5000+ CAT Questions

"Cracku CAT Questions"

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Question 31

Among the students surveyed who supported proposal A, what percentage preferred Sunita for student union president?

cracku

Students in a college are discussing two proposals –
A: a proposal by the authorities to introduce dress code on campus,
 and
B: a proposal by the students to allow multinational food franchises to set up outlets on college campus.

A student does not necessarily support either of the two proposals.

In an upcoming election for student union president, there are two candidates in fray:
Sunita and Ragini. Every student prefers one of the two candidates.

A survey was conducted among the students by picking a sample of 500 students. The following information was noted from this survey:

1. 250 students supported proposal A and 250 students supported proposal B.
2. Among the 200 students who preferred Sunita as student union president, 80% supported proposal A.
3. Among those who preferred Ragini, 30% supported proposal A.
4. 20% of those who supported proposal B preferred Sunita.
5. 40% of those who did not support proposal B preferred Sunita.
6. Every student who preferred Sunita and supported proposal B also supported proposal A.
7. Among those who preferred Ragini, 20% did not support any of the proposals.

$A = 250$
 $S = 200$
 $R = 300$
 $T = 500$



VIDEO SOLUTION

Instructions

1000 patients currently suffering from a disease were selected to study the effectiveness of treatment of four types of medicines – A, B, C and D. These patients were first randomly assigned into two groups of equal size, called treatment group and control group. The patients in the control group were not treated with any of these medicines; instead they were given a dummy medicine, called placebo, containing only sugar and starch. The following information is known about the patients in the treatment group.

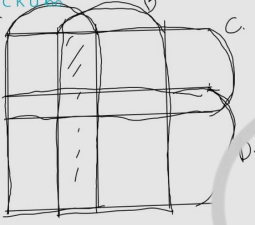
- a. A total of 250 patients were treated with type A medicine and a total of 210 patients were treated with type C medicine.
- b. 25 patients were treated with type A medicine only. 20 patients were treated with type C medicine only. 10 patients were treated with type D medicine only.
- c. 35 patients were treated with type A and type D medicines only. 20 patients were treated with type A and type B medicines only. 30 patients were treated with type A and type C medicines only. 20 patients were treated with type C and type D medicines only.
- d. 100 patients were treated with exactly three types of medicines.
- e. 40 patients were treated with medicines of types A, B and C, but not with medicines of type D. 20 patients

were treated with medicines of types A, C and D, but not with medicines of type B.

f. 50 patients were given all the four types of medicines. 75 patients were treated with exactly one type of medicine.

Question 32

The number of patients who were treated with medicine type D was:



1000 patients currently suffering from a disease were selected to study the effectiveness of treatment of four types of medicines – A, B, C and D. These patients were first randomly assigned into two groups of equal size, called treatment group and control group. The patients in the control group were not treated with any of these medicines; instead they were given a dummy medicine, called placebo, containing only sugar and starch. The following information is known about the treatment group.

a. A total of 250 patients were treated with type A medicine and a total of 210 patients were treated with type C medicine.

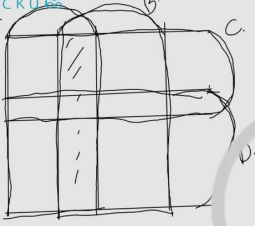
b. 25 patients were treated with type A medicine only, 20 patients were treated with type C medicine only, 10 patients were treated with type D medicine only.

c. 40 patients were treated with type A and type D medicines only, 20 patients were treated with type C and type D medicines only, 30 patients were treated with type A medicines only, 30 patients were treated with type C and type D medicines only.

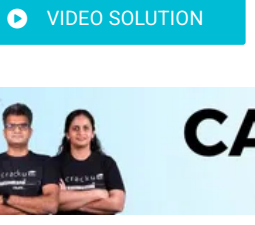
d. 40 patients were treated with exactly three types of medicines.

e. 40 patients were treated with medicines of types A, B and C, but not with medicines of type D. 50 patients were treated with medicines of types A, C and D, but not with medicine type B.

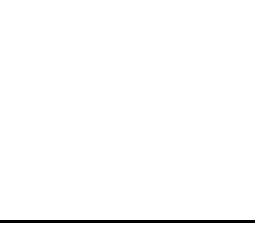
f. 50 patients were given all the four types of medicines. 75 patients were treated with exactly one type of medicine.



500 (500)



500 (500)

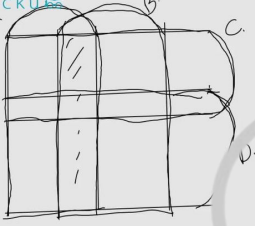


500 (500)

VIDEO SOLUTION

Question 33

How many patients were treated with medicine type B?



1000 patients currently suffering from a disease were selected to study the effectiveness of treatment of four types of medicines – A, B, C and D. These patients were first randomly assigned into two groups of equal size, called treatment group and control group. The patients in the control group were not treated with any of these medicines; instead they were given a dummy medicine, called placebo, containing only sugar and starch. The following information is known about the treatment group.

a. A total of 250 patients were treated with type A medicine and a total of 210 patients were treated with type C medicine.

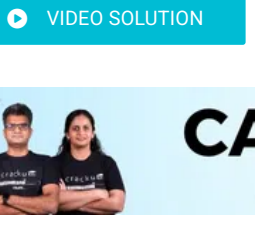
b. 25 patients were treated with type A medicine only, 20 patients were treated with type C medicine only, 10 patients were treated with type D medicine only.

c. 40 patients were treated with type A and type D medicines only, 20 patients were treated with type C and type D medicines only, 30 patients were treated with type A medicines only, 30 patients were treated with type C and type D medicines only.

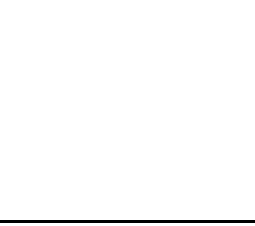
d. 40 patients were treated with exactly three types of medicines.

e. 40 patients were treated with medicines of types A, B and C, but not with medicines of type D. 50 patients were treated with medicines of types A, C and D, but not with medicine type B.

f. 50 patients were given all the four types of medicines. 75 patients were treated with exactly one type of medicine.



500 (500)



500 (500)

VIDEO SOLUTION



CAT Daily Targets

Step by step, heading towards 99+ percentile

Question 34

The number of patients who were treated with medicine types B, C and D, but not type A was:

cracku

1000 patients currently suffering from a disease were selected to study the effectiveness of treatment of four types of medicines – A, B, C and D. These patients were first randomly assigned into two groups of equal size, called treatment group and control group. The patients in the control group were not treated with any of these medicines; instead they were given a dummy medicine, called placebo, containing only sugar and starch. The following information is known about the treatment group.

a. A total of 250 patients were treated with type A medicine and a total of 210 patients were treated with type C medicine.
 b. 25 patients were treated with type A medicine only, 20 patients were treated with type C medicine only, 10 patients were treated with type D medicine only.
 c. 40 patients were treated with type A and type D medicines only, 20 patients were treated with type C and type D medicines only, 30 patients were treated with type A and type D medicines only.
 d. 40 patients were treated with medicines of types A, B and C, but not with medicines of type D. 20 patients were treated with medicines of types A, C and D, but not with medicines of type B.
 e. 50 patients were given all the four types of medicines. 75 patients were treated with only one type of medicine.

VIDEO SOLUTION

Question 35

How many patients were treated with medicine types B and D only?

cracku

1000 patients currently suffering from a disease were selected to study the effectiveness of treatment of four types of medicines – A, B, C and D. These patients were first randomly assigned into two groups of equal size, called treatment group and control group. The patients in the control group were not treated with any of these medicines; instead they were given a dummy medicine, called placebo, containing only sugar and starch. The following information is known about the treatment group.

a. A total of 250 patients were treated with type A medicine and a total of 210 patients were treated with type C medicine.
 b. 25 patients were treated with type A medicine only, 20 patients were treated with type C medicine only, 10 patients were treated with type D medicine only.
 c. 40 patients were treated with type A and type D medicines only, 20 patients were treated with type C and type D medicines only, 30 patients were treated with type A and type D medicines only.
 d. 40 patients were treated with medicines of types A, B and C, but not with medicines of type D. 20 patients were treated with medicines of types A, C and D, but not with medicines of type B.
 e. 50 patients were given all the four types of medicines. 75 patients were treated with only one type of medicine.

VIDEO SOLUTION

Instructions

Ten musicians (A, B, C, D, E, F, G, H, I and J) are experts in at least one of the following three percussion instruments: tabla, mridangam, and ghatam. Among them, three are experts in tabla but not in mridangam or ghatam, another three are experts in mridangam but not in tabla or ghatam, and one is an expert in ghatam but not in tabla or mridangam. Further, two are experts in tabla and mridangam but not in ghatam, and one is an expert in tabla and ghatam but not in mridangam.

The following facts are known about these ten musicians.

- Both A and B are experts in mridangam, but only one of them is also an expert in tabla.
- D is an expert in both tabla and ghatam.
- Both F and G are experts in tabla, but only one of them is also an expert in mridangam.
- Neither I nor J is an expert in tabla.
- Neither H nor I is an expert in mridangam, but only one of them is an expert in ghatam.

Question 36

Who among the following is DEFINITELY an expert in tabla but not in either mridangam or ghatam?

- A F
- B C
- C A

D H

The video solution for Question 37 displays a Venn diagram with three overlapping circles labeled 'Tabla', 'Mrid', and 'Ghatam'. The regions are numbered 1 through 6. The 'Tabla' circle contains 3, the 'Mrid' circle contains 2, and the 'Ghatam' circle contains 1. The intersection of 'Tabla' and 'Mrid' contains 2, the intersection of 'Tabla' and 'Ghatam' contains 1, and the intersection of 'Mrid' and 'Ghatam' contains 1. The intersection of all three contains 1. The video also shows a list of facts and a list of options.

The following facts are known about these ten musicians.

1. Both A and B are experts in mridangam, but only one of them is also an expert in tabla.
2. D is an expert in both tabla and ghatam.
3. Both F and G are experts in tabla, but only one of them is also an expert in mridangam.
4. Neither I nor J is an expert in tabla.
5. Neither H nor I is an expert in mridangam, but only one of them is an expert in ghatam.

A, B =

VIDEO SOLUTION

FREE

CAT Sectional Tests

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Question 37

Which of the following pairs CANNOT have any musician who is an expert in both tabla and mridangam but not in ghatam?

- A F and G
- B C and E
- C A and B
- D C and F

The video solution for Question 37 displays a Venn diagram with three overlapping circles labeled 'Tabla', 'Mrid', and 'Ghatam'. The regions are numbered 1 through 6. The 'Tabla' circle contains 3, the 'Mrid' circle contains 2, and the 'Ghatam' circle contains 1. The intersection of 'Tabla' and 'Mrid' contains 2, the intersection of 'Tabla' and 'Ghatam' contains 1, and the intersection of 'Mrid' and 'Ghatam' contains 1. The intersection of all three contains 1. The video also shows a list of facts and a list of options.

The following facts are known about these ten musicians.

1. Both A and B are experts in mridangam, but only one of them is also an expert in tabla.
2. D is an expert in both tabla and ghatam.
3. Both F and G are experts in tabla, but only one of them is also an expert in mridangam.
4. Neither I nor J is an expert in tabla.
5. Neither H nor I is an expert in mridangam, but only one of them is an expert in ghatam.

A, B =

VIDEO SOLUTION

Question 38

If C is an expert in mridangam and F is not, then which are the three musicians who are experts in tabla but not in either mridangam or ghatam?

- A E, F and H

- B** C, G and H
- C** E, G and H
- D** C, E and G

cracku

Table

Mrid

Ghatam

The following facts are known about these ten musicians.

- Both A and B are experts in mridangam, but only one of them is also an expert in tabla.
- D is an expert in both tabla and ghatam.
- Both F and G are experts in tabla, but only one of them is also an expert in mridangam.
- Neither I nor J is an expert in tabla.
- Neither H nor I is an expert in mridangam, but only one of them is an expert in ghatam.

A, B =

VIDEO SOLUTION

Question 39

Who among the following is DEFINITELY an expert in mridangam but not in either tabla or ghatam?

- A** B
- B** J
- C** G
- D** E

cracku

Table

Mrid

Ghatam

The following facts are known about these ten musicians.

- Both A and B are experts in mridangam, but only one of them is also an expert in tabla.
- D is an expert in both tabla and ghatam.
- Both F and G are experts in tabla, but only one of them is also an expert in mridangam.
- Neither I nor J is an expert in tabla.
- Neither H nor I is an expert in mridangam, but only one of them is an expert in ghatam.

A, B =

VIDEO SOLUTION

cracku

CAT Study Material

All CAT Free Resources

Instructions

A speciality supermarket sells 320 products. Each of these products was either a cosmetic product or a nutrition product. Each of these products was also either a foreign product or a domestic product. Each of these products had at least one of the two approvals - FDA or EU.

1. There were equal numbers of domestic and foreign products.
2. Half of the domestic products were FDA approved cosmetic products.
3. None of the foreign products had both the approvals, while 60 domestic products had both the approvals.
4. There were 140 nutrition products, half of them were foreign products.
5. There were 200 FDA approved products. 70 of them were foreign products and 120 of them were cosmetic products.

How many foreign products were FDA approved cosmetic products?

 VIDEO SOLUTION

How many cosmetic products did not have FDA approval?

- VIDEO SOLUTION

If 70 cosmetic products did not have EU approval, then how many nutrition products had both the approvals?

- 22/95

B 30

C 10

D 20



A speciality supermarket sells 320 products. Each of these products was either a cosmetic product or a nutrition product. Each of these products was also either a foreign product or a domestic product. Each of these products had at least one of the two approvals - FDA or EU.

The following facts are also known:

1. There were equal numbers of domestic and foreign products.
2. Half of the domestic products were FDA approved cosmetic products.
3. None of the foreign products had both the approvals, while 60 domestic products had both the approvals.
4. There were 140 nutrition products, half of them were foreign products.
5. There were 200 FDA approved products. 70 of them were foreign products and 130 were domestic products.

Handwritten solution diagram:

```
graph TD
    Total[320] --> C[Cosmetic]
    Total --> N[Nutrition]
    C --> F[Cosmetic Foreign]
    C --> D[Cosmetic Domestic]
    N --> F[Nutrition Foreign]
    N --> D[Nutrition Domestic]
    F --> FDA[140]
    F --> EU[140]
    D --> FDA[130]
    D --> EU[130]
    FDA --> FDA_EU[130]
    FDA --> FDA_N[100]
    EU --> FDA_EU[130]
    EU --> EU_N[100]
    FDA_EU --> FDA_EU_C[60]
    FDA_EU --> FDA_EU_N[70]
    FDA_N --> FDA_N_C[10]
    FDA_N --> FDA_N_N[90]
    EU_N --> EU_N_C[10]
    EU_N --> EU_N_N[90]
```

VIDEO SOLUTION



Question 43

If 50 nutrition products did not have EU approval, then how many domestic cosmetic products did not have EU approval?



A speciality supermarket sells 320 products. Each of these products was either a cosmetic product or a nutrition product. Each of these products was also either a foreign product or a domestic product. Each of these products had at least one of the two approvals - FDA or EU.

The following facts are also known:

1. There were equal numbers of domestic and foreign products.
2. Half of the domestic products were FDA approved cosmetic products.
3. None of the foreign products had both the approvals, while 60 domestic products had both the approvals.
4. There were 140 nutrition products, half of them were foreign products.
5. There were 200 FDA approved products. 70 of them were foreign products and 130 were domestic products.

Handwritten solution diagram:

```
graph TD
    Total[320] --> C[Cosmetic]
    Total --> N[Nutrition]
    C --> F[Cosmetic Foreign]
    C --> D[Cosmetic Domestic]
    N --> F[Nutrition Foreign]
    N --> D[Nutrition Domestic]
    F --> FDA[140]
    F --> EU[140]
    D --> FDA[130]
    D --> EU[130]
    FDA --> FDA_EU[130]
    FDA --> FDA_N[100]
    EU --> FDA_EU[130]
    EU --> EU_N[100]
    FDA_EU --> FDA_EU_C[60]
    FDA_EU --> FDA_EU_N[70]
    FDA_N --> FDA_N_C[10]
    FDA_N --> FDA_N_N[90]
    EU_N --> EU_N_C[10]
    EU_N --> EU_N_N[90]
```

VIDEO SOLUTION

Question 44

Which among the following options best represents the number of domestic cosmetic products that had both the approvals?

A At least 10 and at most 60

B At least 10 and at most 80

C At least 20 and at most 70

D At least 20 and at most 50

VIDEO SOLUTION

Instructions

A survey of 600 schools in India was conducted to gather information about their online teaching learning processes (OTLP). The following four facilities were studied.

- F1: Own software for OTLP
- F2: Trained teachers for OTLP
- F3: Training materials for OTLP
- F4: All students having Laptops

The following observations were summarized from the survey.

1. 80 schools did not have any of the four facilities - F1, F2, F3, F4.
2. 40 schools had all four facilities.
3. The number of schools with only F1, only F2, only F3, and only F4 was 25, 30, 26 and 20 respectively.
4. The number of schools with exactly three of the facilities was the same irrespective of which three were considered.
5. 313 schools had F2.
6. 26 schools had only F2 and F3 (but neither F1 nor F4).
7. Among the schools having F4, 24 had only F3, and 45 had only F2.
8. 162 schools had both F1 and F2.
9. The number of schools having F1 was the same as the number of schools having F4.

Question 45

What was the number of schools having only facilities F1 and F4?

VIDEO SOLUTION

100+ CAT Quant Questions



Question 46

What was the number of schools having facilities F2 and F4?

- A 185
- B 95
- C 45
- D 85

A survey of 600 schools in India was conducted to gather information about their online teaching learning processes (OTLP). The following four facilities were studied.

E1: Own software for OTLP
F2: Trained teachers for OTLP
F3: Materials for OTLP
F4: All students having Laptops

The following observations were summarized from the survey:

- 10 schools did not have any of the four facilities - F1, F2, F3, F4.
- 25 schools had all four facilities.
- 25 schools had only F1, only F2, only F3, and only F4 was 25.
- 5 schools had exactly two of the facilities.
- 5 schools had exactly three of the facilities was the same.
- 5 schools had F1.
- 26 schools had F2 and F3 (but neither F1 nor F4).
7. Among the schools having F4, 24 had only F3, and 45 had only F2.
8. 10 schools had both F1 and F2.
9. The number of schools having F1 was the same as the number having F4.

VIDEO SOLUTION

Question 47

What was the total number of schools having exactly three of the four facilities?

- A 64
- B 50
- C 200
- D 80

A survey of 600 schools in India was conducted to gather information about their online teaching learning processes (OTLP). The following four facilities were studied.

E1: Own software for OTLP
F2: Trained teachers for OTLP
F3: Materials for OTLP
F4: All students having Laptops

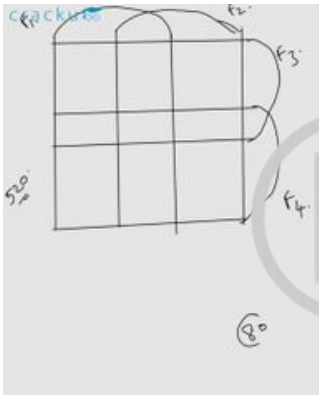
The following observations were summarized from the survey:

- 10 schools did not have any of the four facilities - F1, F2, F3, F4.
- 25 schools had all four facilities.
- 25 schools had only F1, only F2, only F3, and only F4 was 25.
- 5 schools had exactly two of the facilities.
- 5 schools had exactly three of the facilities was the same.
- 5 schools had F1.
- 26 schools had F2 and F3 (but neither F1 nor F4).
7. Among the schools having F4, 24 had only F3, and 45 had only F2.
8. 10 schools had both F1 and F2.
9. The number of schools having F1 was the same as the number having F4.

VIDEO SOLUTION

Question 48

What was the number of schools having only facilities F1 and F3?



A survey of 600 schools in India was conducted to gather information about their online teaching-learning processes (OTLP). The following four facilities were studied:

- F1: Own software for OTLP
- F2: Trained teachers for OTLP
- F3: Trained materials for OTLP
- F4: All students having Laptops

The following observations were summarized from the survey:

- 50 schools did not have any of the four facilities - F1, F2, F3, F4.
- 80 schools had all four facilities.
- 25 schools had only F1, only F2, only F3, and only F4 was 25.
- The number of schools with exactly three of the facilities was the same for each of the four facilities.
- 25 schools had F1 and F2 (but neither F3 nor F4).
- 25 schools had F1 and F3 (but neither F2 nor F4).
- 25 schools had F1 and F4 (but neither F2 nor F3).
- 25 schools had F2 and F3 (but neither F1 nor F4).
- 25 schools had F2 and F4 (but neither F1 nor F3).
- 25 schools had F3 and F4 (but neither F1 nor F2).
- 25 schools had all four facilities.

VIDEO SOLUTION

100+ CAT DILR Questions



Answers

1.C	2.A	3.B	4.D	5.C	6.D	7.C	8.C
9.D	10.B	11.60	12.170	13.A	14.A	15.A	16.C
17.A	18.C	19.2	20.A	21.C	22.4	23.A	24.50
25.C	26.C	27.B	28.A	29.D	30.84	31.64	32.325
33.340	34.10	35.150	36.D	37.B	38.A	39.B	40.40
41.D	42.C	43.50	44.A	45.20	46.A	47.C	48.42

Explanations

1. C

No fishes breathe through lungs but all whales breathe through lungs. So, no whale can be a fish. Therefore, ADE is a valid combination of logically connected statements.
Hence, option C is the correct answer.

[VIEW SOLUTION](#)

2. A

In option A, D can be concluded from C and E.
In option B, A, C and B are not connected.
In option C, C cannot be inferred from B and D.
In option D, E and F are not related.

Hence, option A is the correct answer.

[VIEW SOLUTION](#)

3. B

Everyone who is sane can do logic and all who can do logic are fit to serve on a jury. So, all who can do logic are fit to serve on a jury.
Hence, option B is the correct answer.

[VIEW SOLUTION](#)

4. D

In option A, E is not a valid conclusion from A and B.
In option B, F cannot be inferred from A and C.
In option C, E cannot be concluded from A and D.

In option D, If some learned men are candid but all candid men recognize merit in a rival, it can be concluded that some men recognize merit in a rival are learned.

Hence, option D is the correct answer.

[VIEW SOLUTION](#)

5. C

In option A, A is inconsistent with E. It is not given in A that only those who achieve great ends are happy.
In option B, F contradicts with B.
In option C, B can be inferred from A and C.
In option D, F cannot be concluded from A and D.

Hence, option C is the correct answer.

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100+ CAT VARC Questions



6. D

Nothing uncertain is worth dying for. and all beliefs are uncertain. So, no belief is worth dying for. Hence, option D is the correct answer.

[VIEW SOLUTION](#)

7. C

There are six sentences given. So, there can be many combinations of the sentences possible. Therefore, we will check each of the option to find the correct answer.

In option A, BCD is given. Statement C says all of my closest friends disapprove of me and D says all who disapprove of me are aardvarks. So, all my closest friends must be aardvarks. But, this contradicts with statement B. Thus, option A is incorrect.

In option B, statement D cannot be inferred from A and B.

In option C, all three statements are logically connected.

In option D, E cannot be inferred from A and B.

Hence, option C is the correct answer.

[VIEW SOLUTION](#)

8. C

All students of this college rank as University students and all University students are eligible to enter for the prize. Therefore, all first year students are entitled to enter for the prize.

Hence, option C is the correct answer.

[VIEW SOLUTION](#)

9. D

ADF is false as we cannot infer F(Some mammals are whales.) even if A(Some mammals are carnivores.) and D(All whales are carnivores.) are true.

ABC and AEF all statements are different, the set is not related.

All whales are mammals and all whales are aquatic animals. So, some aquatic animals are mammals. Therefore, BCE is valid combination.

Hence, option D is the correct answer.

[VIEW SOLUTION](#)

10. B

Some men are men of artistic genius and all men are men of scientific ability. Therefore, all men of artistic genius are men of scientific ability.

Thus, ACE is the correct combination.

Hence, option B is the correct answer.

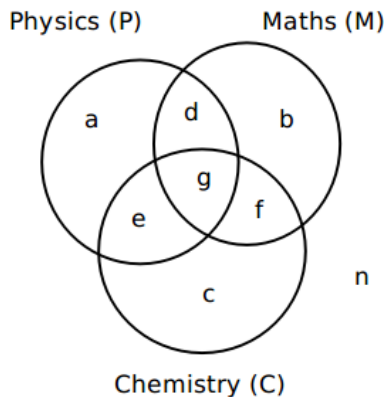
CAT Expected Questions PDF

cracku



11.60

It is given that 200 candidates scored above 90th percentile overall in CET. Let the following Venn diagram represent the number of persons who scored above 80 percentile in CET in each of the three sections:



From (1), $n = 0$

From (2), $d + e + f = 150$

From (3), $a = b = c$

Since, there are a total of 200 candidates

$$a + b + c + g = 200 - 150 = 50$$

$$3a + g = 50 \Rightarrow a < 17$$

$$\text{From (4), } (b + f + c) : (a + d + b) : (a + e + c) = 4 : 2 : 1$$

$$\text{Or } (2a + f) : (2a + d) : (2a + e) = 4 : 2 : 1$$

$$\text{or, } 6a + (d + e + f) = 7x$$

$$\text{Or, } 6a + 150 = 7x$$

So, a can be 3 or 10

x can be 24 or 30

$$2a + e \text{ can be } 24 \text{ or } 30 \Rightarrow e \text{ can be } 18 \text{ or } 10$$

$$2a + d \text{ can be } 48 \text{ or } 60 \Rightarrow d \text{ can be } 42 \text{ or } 40$$

$$2a + f \text{ can be } 96 \text{ or } 120 \Rightarrow f \text{ can be } 90 \text{ or } 100$$

$$3a + g = 50 \Rightarrow g \text{ can be } 41 \text{ or } 20$$

Among the candidates who are at or above 90th percentile, the candidates who are at or above 80th percentile in at least two sections are selected for AET. Hence, the candidates represented by d, e, f and g are selected for AET.

BIE will consider the candidates who are appearing for AET and are at or above 80th percentile in P. Hence, BIE will consider the candidates represented by d, e and g , which can be 104 or 80.

BIE will conduct a separate test for the other students who are at or above 80th percentile in P. Given that there are a total of 400 candidates at or above 80th percentile in P, and since there are 104 or 80 candidates at or above 80th percentile in P and are at or above 90th percentile overall, there must be 296 or 320 candidates at or above 80th percentile in P who scored less than 90th percentile overall.

From the given condition, g is a multiple of 5.

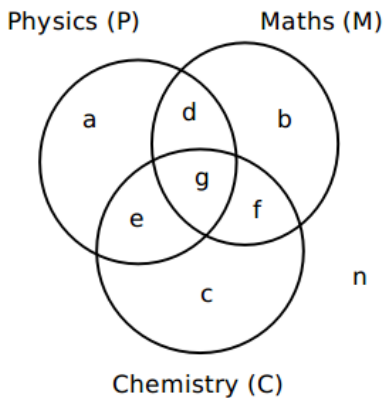
So, $g = 20$.

The number of candidates at or above 90th percentile overall and at or above 80th percentile in both P and M = $d + g = 60$.

Hence, 60 is the correct answer.

12.170

It is given that 200 candidates scored above 90th percentile overall in CET. Let the following Venn diagram represent the number of persons who scored above 80 percentile in CET in each of the three sections:



From (1), $n = 0$

From (2), $d + e + f = 150$

From (3), $a = b = c$

Since, there are a total of 200 candidates

$$a + b + c + g = 200 - 150 = 50$$

$$3a + g = 50 \Rightarrow a < 17$$

$$\text{From (4), } (b + f + c) : (a + d + b) : (a + e + c) = 4 : 2 : 1$$

$$\text{Or } (2a + f) : (2a + d) : (2a + e) = 4 : 2 : 1$$

$$\text{or, } 6a + (d + e + f) = 7x$$

$$\text{Or, } 6a + 150 = 7x$$

So, a can be 3 or 10

x can be 24 or 30

$$2a + e \text{ can be } 24 \text{ or } 30 \Rightarrow e \text{ can be } 18 \text{ or } 10$$

$$2a + d \text{ can be } 48 \text{ or } 60 \Rightarrow d \text{ can be } 42 \text{ or } 40$$

$$2a + f \text{ can be } 96 \text{ or } 120 \Rightarrow f \text{ can be } 90 \text{ or } 100$$

$$3a + g = 50 \Rightarrow g \text{ can be } 41 \text{ or } 20$$

Among the candidates who are at or above 90th percentile, the candidates who are at or above 80th percentile in at least two sections are selected for AET. Hence, the candidates represented by d, e, f and g are selected for AET.

BIE will consider the candidates who are appearing for AET and are at or above 80th percentile in P. Hence, BIE will consider the candidates represented by d, e and g , which can be 104 or 80.

BIE will conduct a separate test for the other students who are at or above 80th percentile in P. Given that there are a total of 400 candidates at or above 80th percentile in P, and since there are 104 or 80 candidates at or above 80th percentile in P and are at or above 90th percentile overall, there must be 296 or 320 candidates at or above 80th percentile in P who scored less than 90th percentile overall.

In this question, $g = 20$.

Number of candidates shortlisted for AET

$$= d + e + f + g$$

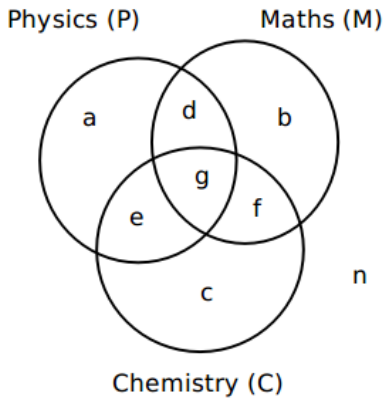
$$= 40 + 10 + 100 + 20$$

$$= 170$$

Hence, 170 is the correct answer.

13.A

It is given that 200 candidates scored above 90th percentile overall in CET. Let the following Venn diagram represent the number of persons who scored above 80 percentile in CET in each of the three sections:



From (1), $n = 0$

From (2), $d + e + f = 150$

From (3), $a = b = c$

Since, there are a total of 200 candidates

$$a + b + c + g = 200 - 150 = 50$$

$$3a + g = 50 \Rightarrow a < 17$$

$$\text{From (4), } (b + f + c) : (a + d + b) : (a + e + c) = 4 : 2 : 1$$

$$\text{Or } (2a + f) : (2a + d) : (2a + e) = 4 : 2 : 1$$

$$\text{or, } 6a + (d + e + f) = 7x$$

$$\text{Or, } 6a + 150 = 7x$$

So, a can be 3 or 10

x can be 24 or 30

$$2a + e \text{ can be } 24 \text{ or } 30 \Rightarrow e \text{ can be } 18 \text{ or } 10$$

$$2a + d \text{ can be } 48 \text{ or } 60 \Rightarrow d \text{ can be } 42 \text{ or } 40$$

$$2a + f \text{ can be } 96 \text{ or } 120 \Rightarrow f \text{ can be } 90 \text{ or } 100$$

$$3a + g = 50 \Rightarrow g \text{ can be } 41 \text{ or } 20$$

Among the candidates who are at or above 90th percentile, the candidates who are at or above 80th percentile in at least two sections are selected for AET. Hence, the candidates represented by d , e , f and g are selected for AET.

BIE will consider the candidates who are appearing for AET and are at or above 80th percentile in P. Hence, BIE will consider the candidates represented by d , e and g , which can be 104 or 80.

BIE will conduct a separate test for the other students who are at or above 80th percentile in P. Given that there are a total of 400 candidates at or above 80th percentile in P, and since there are 104 or 80 candidates at or above 80th percentile in P and are at or above 90th percentile overall, there must be 296 or 320 candidates at or above 80th percentile in P who scored less than 90th percentile overall.

From the given condition, the number of candidates at or above 90th percentile overall and at or above 80th percentile in P in CET = $(3 + 18 + 42 + 41) = 104$.

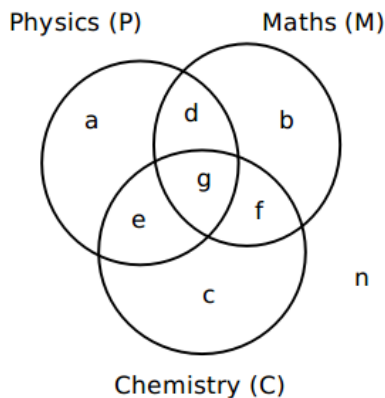
The number of candidates who have to sit for separate test = $(400 - 104 + 3) = 296 + 3 = 299$ (we have added 3 for those who have scored more than 80th percentile only in P which is ' a ').

Hence, option A is the correct answer.

[▶ VIDEO SOLUTION](#)

14. A

It is given that 200 candidates scored above 90th percentile overall in CET. Let the following Venn diagram represent the number of persons who scored above 80 percentile in CET in each of the three sections:



From (1), $n = 0$

From (2), $d + e + f = 150$

From (3), $a = b = c$

Since, there are a total of 200 candidates

$$a + b + c + g = 200 - 150 = 50$$

$$3a + g = 50 \Rightarrow a < 17$$

$$\text{From (4), } (b + f + c) : (a + d + b) : (a + e + c) = 4 : 2 : 1$$

$$\text{Or } (2a + f) : (2a + d) : (2a + e) = 4 : 2 : 1$$

$$\text{or, } 6a + (d + e + f) = 7x$$

$$\text{Or, } 6a + 150 = 7x$$

So, a can be 3 or 10

x can be 24 or 30

$$2a + e \text{ can be } 24 \text{ or } 30 \Rightarrow e \text{ can be } 18 \text{ or } 10$$

$$2a + d \text{ can be } 48 \text{ or } 60 \Rightarrow d \text{ can be } 42 \text{ or } 40$$

$$2a + f \text{ can be } 96 \text{ or } 120 \Rightarrow f \text{ can be } 90 \text{ or } 100$$

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Among the candidates who are at or above 90th percentile, the candidates who are at or above 80th percentile in at least two sections are selected for AET. Hence, the candidates represented by d, e, f and g are selected for AET.

BIE will consider the candidates who are appearing for AET and are at or above 80th percentile in P. Hence, BIE will consider the candidates represented by d, e and g , which can be 104 or 80.

BIE will conduct a separate test for the other students who are at or above 80th percentile in P. Given that there are a total of 400 candidates at or above 80th percentile in P, and since there are 104 or 80 candidates at or above 80th percentile in P and are at or above 90th percentile overall, there must be 296 or 320 candidates at or above 80th percentile in P who scored less than 90th percentile overall.

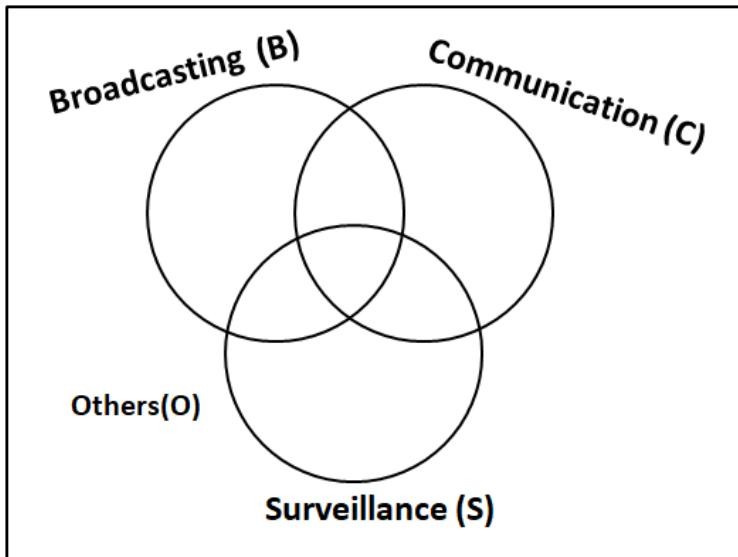
a can be 3 or 10

Hence, Option A is the correct answer.

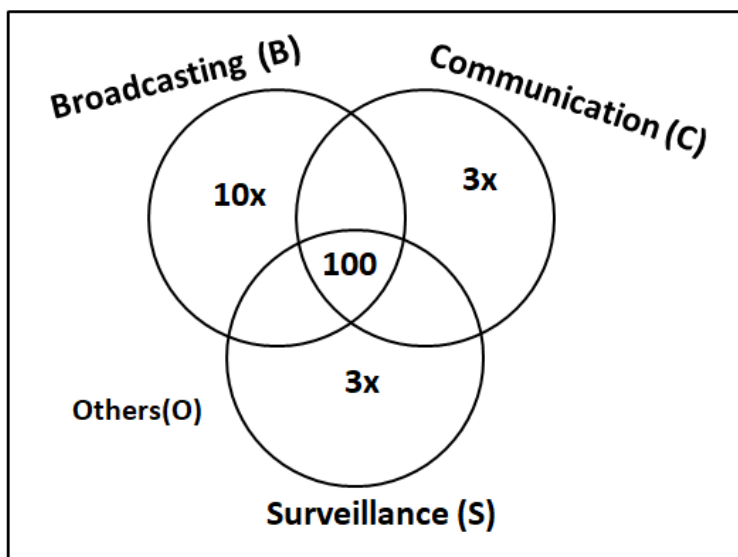
VIDEO SOLUTION

15.A

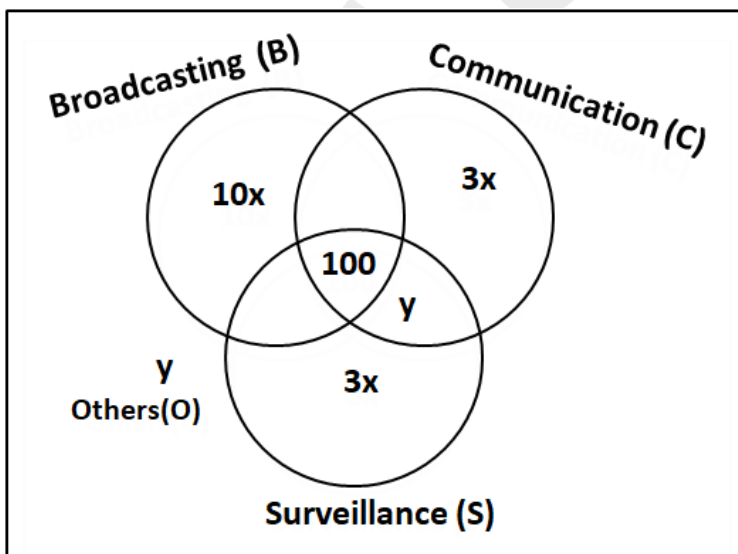
It is given that a satellite serving either B, or C, or S does not serve O. So we can say that it's basically 3 satellites broadcasting (B), communication (C), surveillance (S) which can have intersections. Those satellites which are not part of any category are placed in others. We can draw the Venn diagram as follows.



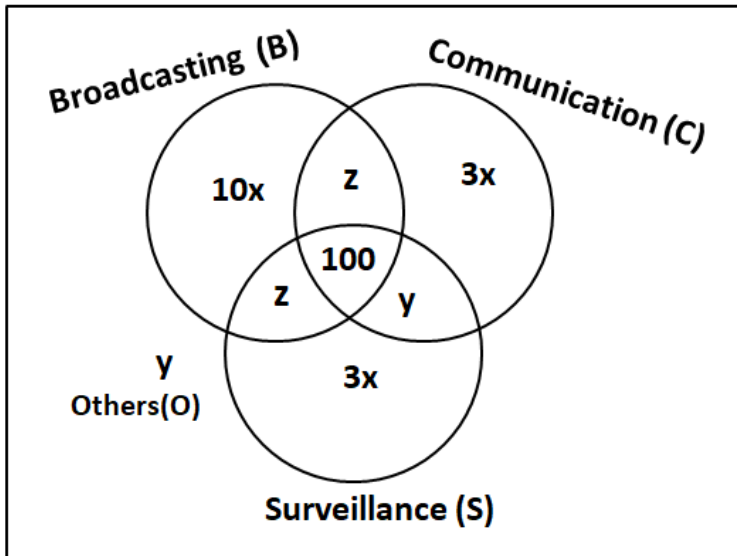
1. The numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1.
 2. The number of satellites serving all three of B, C, and S is 100.
 3. The number of satellites exclusively serving C is the same as the number of satellites exclusively serving S. This number is 30% of the number of satellites exclusively serving B.
 4. The number of satellites serving O is the same as the number of satellites serving both C and S but not B.
- Let '10x' be the number of satellites exclusively serving B. Then, the number of satellites exclusively serving C and S = $0.30 \times 10x = 3x$



Let 'y' be the number of satellites serving others(O).



Let 'z' be the number of satellites serving B, C but not S. Since the numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1. Therefore, we can say that number of satellites serving B, S but not C = z.



It is given that

$$\Rightarrow 10x + 2z + 2y + 6x + 100 = 1600$$

$$\Rightarrow 8x + z + y = 750 \dots (1)$$

The numbers of satellites serving B, C, and S (though maybe not exclusively) are in the ratio 2:1:1.

$$\Rightarrow \frac{10x + 2z + 100}{z + 100 + 3x + y} = \frac{2}{1}$$

$$\Rightarrow 10x + 2z + 100 = 2(z + 100 + 3x + y)$$

$$\Rightarrow 4x = 100 + 2y$$

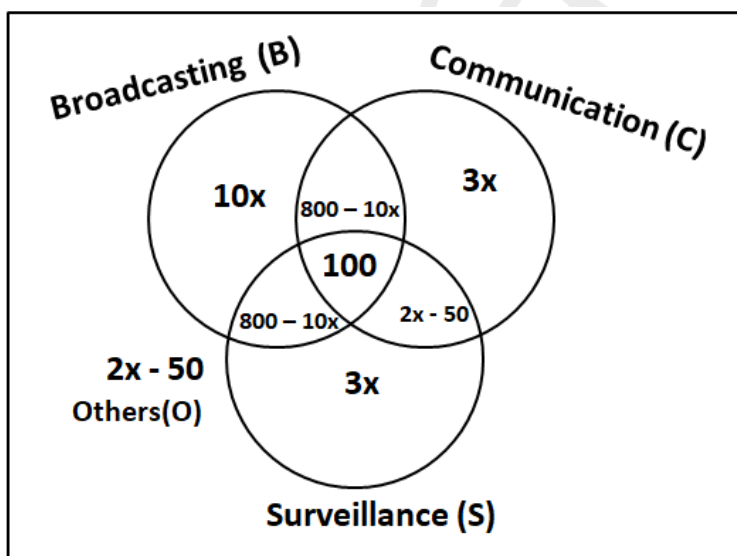
$$\Rightarrow 2x = 50 + y$$

$$\Rightarrow y = 2x - 50 \dots (2)$$

We can substitute this in equation (1)

$$\Rightarrow 8x + z + 2x - 50 = 750$$

$$\Rightarrow z = 800 - 10x \dots (3)$$



Let us define boundary condition for x,

$$\Rightarrow 2x - 50 \geq 0$$

$$\Rightarrow x \geq 25$$

$$\text{Also, } 800 - 10x \geq 0$$

$$\Rightarrow x \leq 80$$

Therefore, we can say that $x \in [25, 80]$.

It is given that at least 100 of the 1600 satellites were serving O.

$$\Rightarrow 2x - 50 \geq 100$$

$$\Rightarrow x \geq 75$$

The number of satellites serving S = $100 + 800 - 10x + 2x - 50 + 3x = 850 - 5x$

At $x_{min} = 75$, the number of satellites serving S = $850 - 5 \cdot 75 = 475$

At $x_{max} = 80$, the number of satellites serving S = $850 - 5 \cdot 80 = 450$

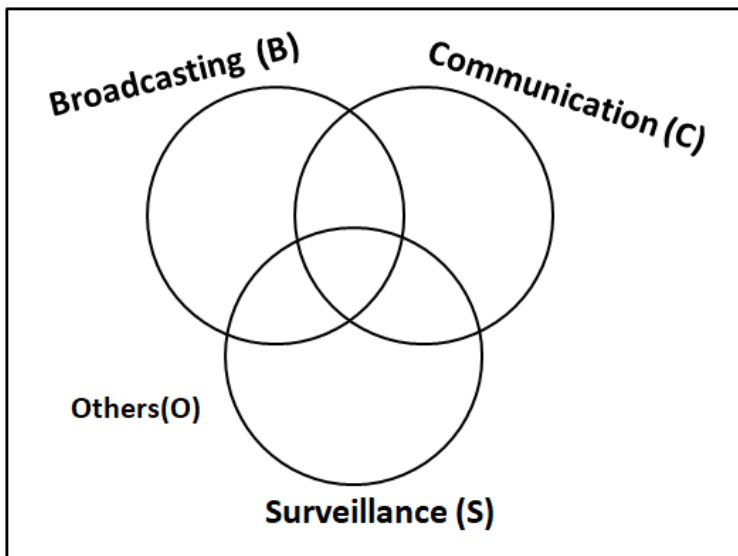
Hence, we can say that the number of satellites serving S must be from 425 to 475. Therefore, we can say that option A is the correct answer.

 VIDEO SOLUTION



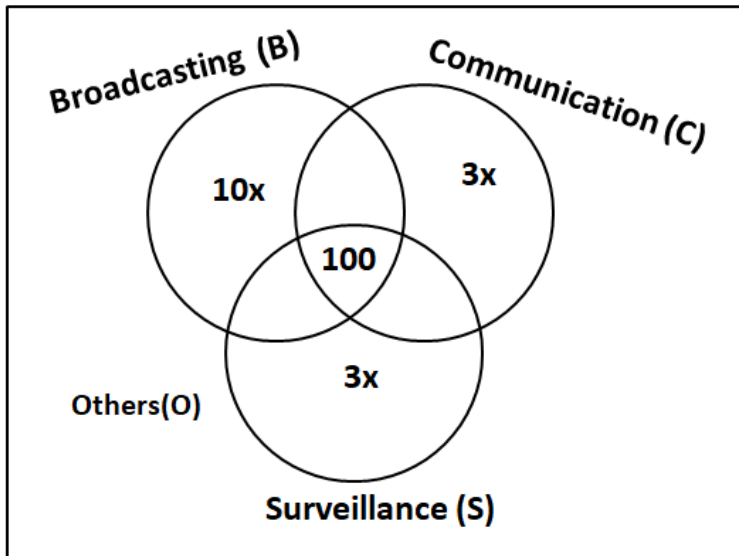
16. C

It is given that a satellite serving either B, or C, or S does not serve O. So we can say that it's basically 3 satellites broadcasting (B), communication (C), surveillance (S) which can have intersections. Those satellites which are not part of any category are placed in others. We can draw the Venn diagram as follows.

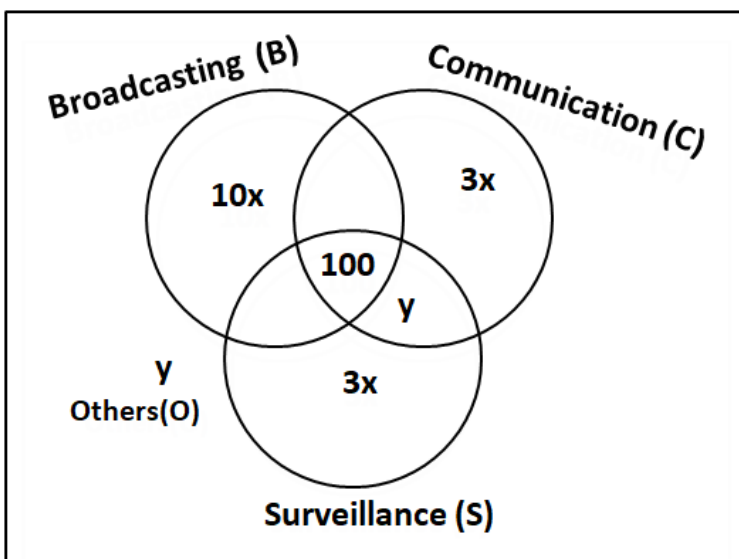


1. The numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1.
2. The number of satellites serving all three of B, C, and S is 100.
3. The number of satellites exclusively serving C is the same as the number of satellites exclusively serving S. This number is 30% of the number of satellites exclusively serving B.
4. The number of satellites serving O is the same as the number of satellites serving both C and S but not B.

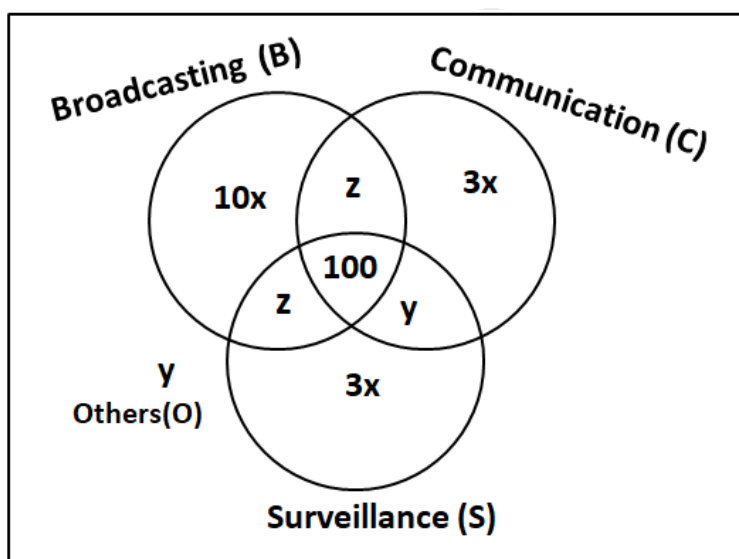
Let '10x' be the number of satellites exclusively serving B. Then, the number of satellites exclusively serving C and S = $0.30 \cdot 10x = 3x$



Let 'y' be the number of satellites serving others(O).



Let 'z' be the number of satellites serving B, C but not S. Since the numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1. Therefore, we can say that number of satellites serving B, S but not C = z.



It is given that

$$\Rightarrow 10x + 2z + 2y + 6x + 100 = 1600$$

$$\Rightarrow 8x + z + y = 750 \dots (1)$$

The numbers of satellites serving B, C, and S (though maybe not exclusively) are in the ratio 2:1:1.

$$\Rightarrow \frac{10x + 2z + 100}{z + 100 + 3x + y} = \frac{2}{1}$$

$$\Rightarrow 10x + 2z + 100 = 2(z + 100 + 3x + y)$$

$$\Rightarrow 4x = 100 + 2y$$

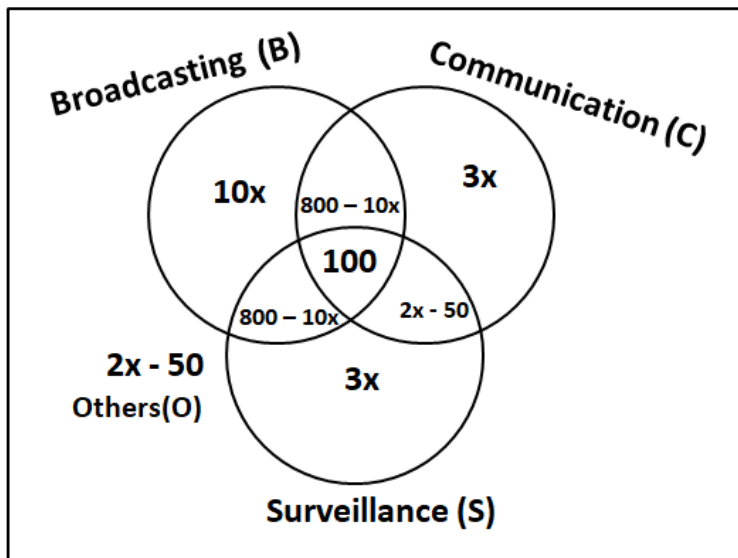
$$\Rightarrow 2x = 50 + y$$

$$\Rightarrow y = 2x - 50 \dots (2)$$

We can substitute this in equation (1)

$$\Rightarrow 8x + z + 2x - 50 = 750$$

$$\Rightarrow z = 800 - 10x \dots (3)$$



Let us define boundary condition for x ,

$$\Rightarrow 2x - 50 \geq 0$$

$$\Rightarrow x \geq 25$$

$$\text{Also, } 800 - 10x \geq 0$$

$$\Rightarrow x \leq 80$$

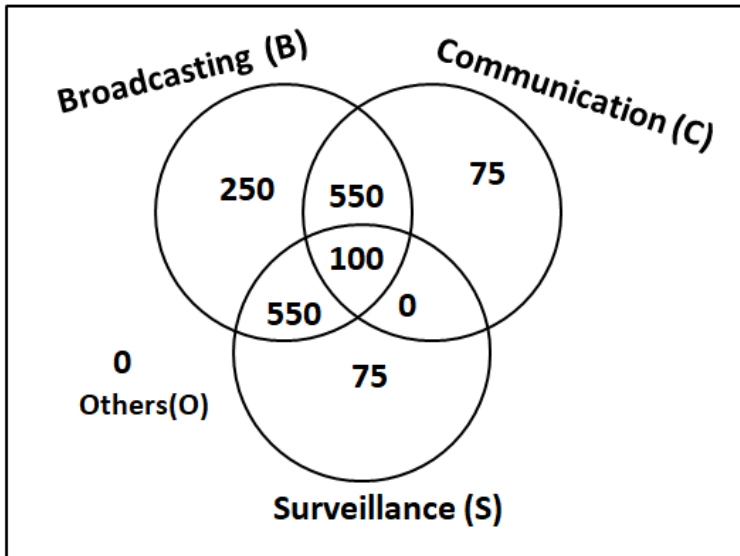
Therefore, we can say that $x \in [25, 80]$.

It is given that the number of satellites serving at least two among B, C, and S is 1200.

$$\Rightarrow 800 - 10x + 800 - 10x + 2x - 50 + 100 = 1200$$

$$\Rightarrow 18x = 450$$

$$\Rightarrow x = 25$$

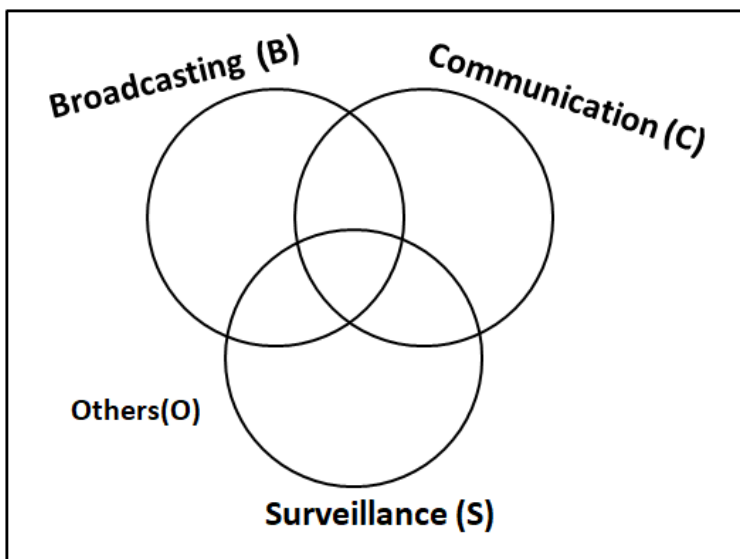


We can determine number of satellites in each of the following category. Hence, option C is definitely false. Therefore, we can say that option C is incorrect.

[VIDEO SOLUTION](#)

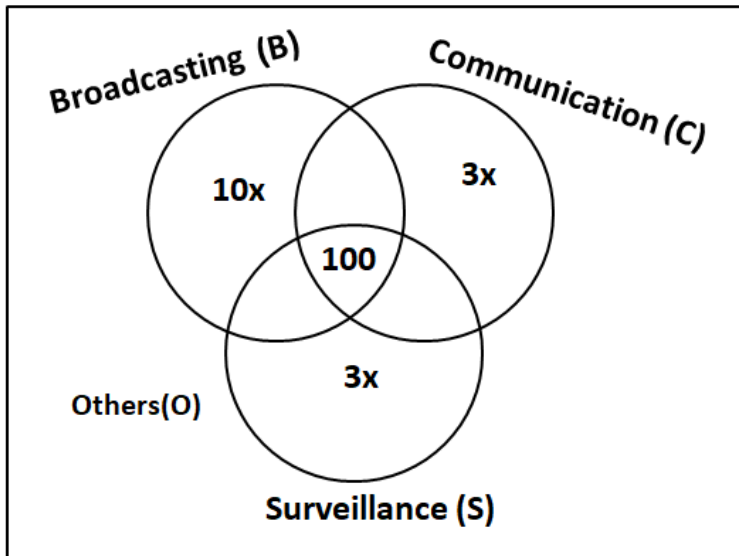
17.A

It is given that a satellite serving either B, or C, or S does not serve O. So we can say that it's basically 3 satellites broadcasting (B), communication (C), surveillance (S) which can have intersections. Those satellites which are not part of any category are placed in others. We can draw the Venn diagram as follows.

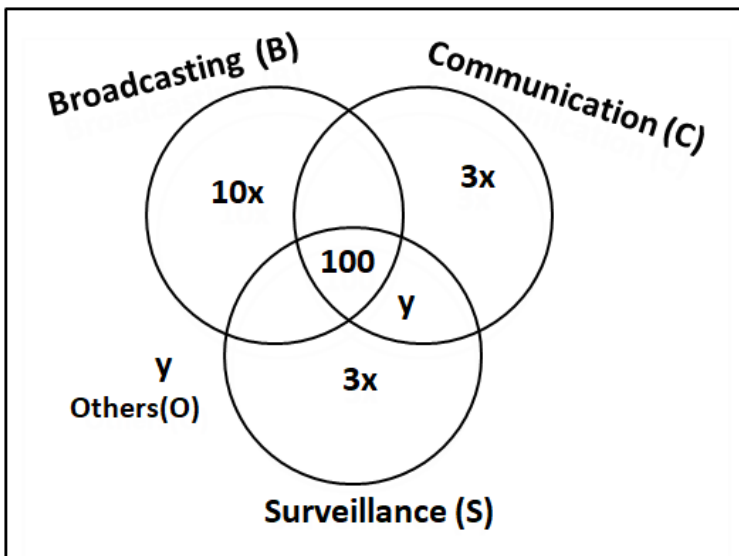


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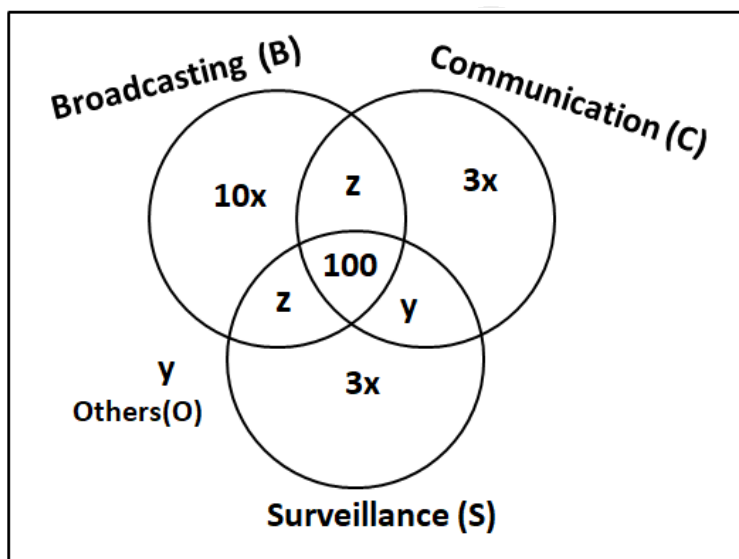
Let '10x' be the number of satellites exclusively serving B. Then, the number of satellites exclusively serving C and S = $0.30 \times 10x = 3x$



Let 'y' be the number of satellites serving others(O).



Let 'z' be the number of satellites serving B, C but not S. Since the numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1. Therefore, we can say that number of satellites serving B, S but not C = z.



It is given that

$$\Rightarrow 10x + 2z + 2y + 6x + 100 = 1600$$

$$\Rightarrow 8x + z + y = 750 \dots (1)$$

The numbers of satellites serving B, C, and S (though maybe not exclusively) are in the ratio 2:1:1.

$$\Rightarrow \frac{10x + 2z + 100}{z + 100 + 3x + y} = \frac{2}{1}$$

$$\Rightarrow 10x + 2z + 100 = 2(z + 100 + 3x + y)$$

$$\Rightarrow 4x = 100 + 2y$$

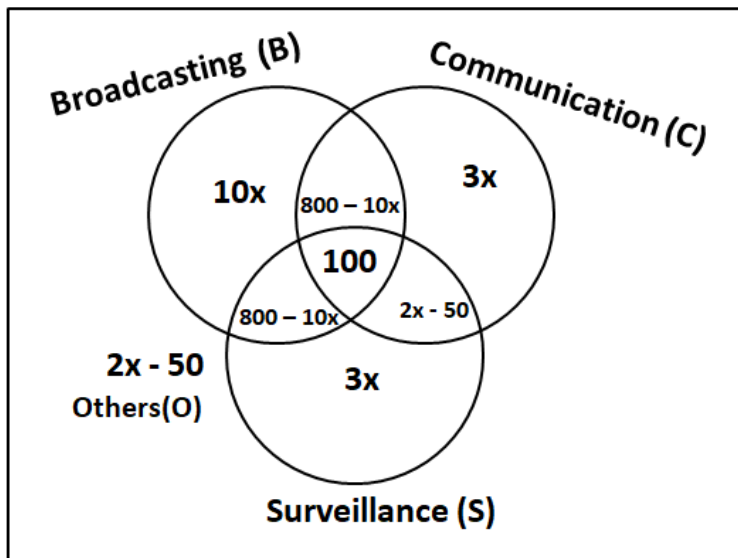
$$\Rightarrow 2x = 50 + y$$

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$$\Rightarrow 8x + z + 2x - 50 = 750$$

$$\Rightarrow z = 800 - 10x \dots (3)$$



Let us define boundary condition for x,

$$\Rightarrow 2x - 50 \geq 0$$

$$\Rightarrow x \geq 25$$

$$\text{Also, } 800 - 10x \geq 0$$

$$\Rightarrow x \leq 80$$

Therefore, we can say that $x \in [25, 80]$.

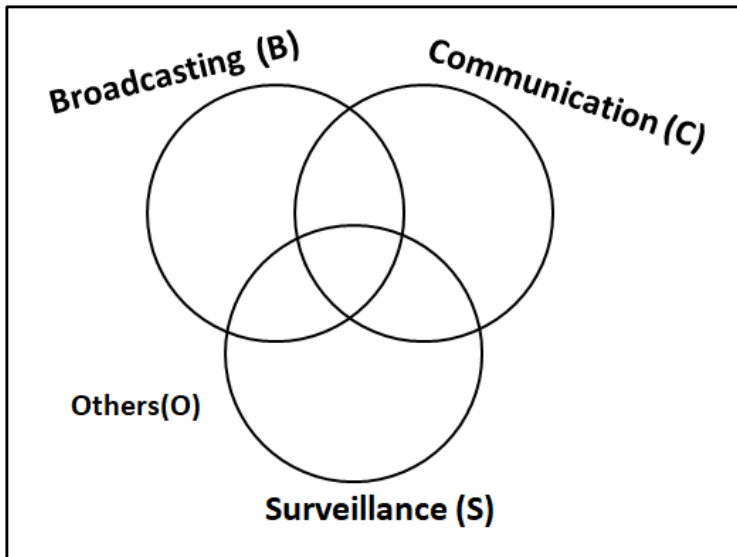
The number of satellites serving B exclusively = $10x$. This will be the minimum when 'x' is the minimum.

At $x_{min} = 25$, The number of satellites serving B exclusively = $10 \times 25 = 250$. Hence, option A is the correct answer.

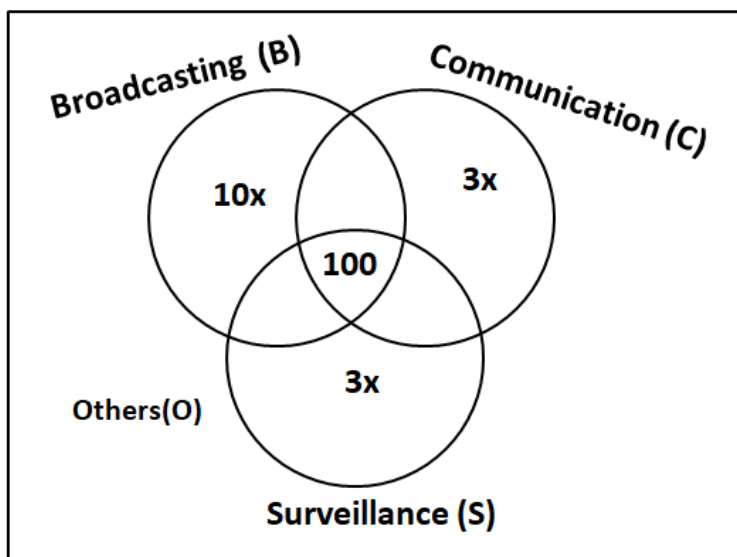
[VIDEO SOLUTION](#)

18. C

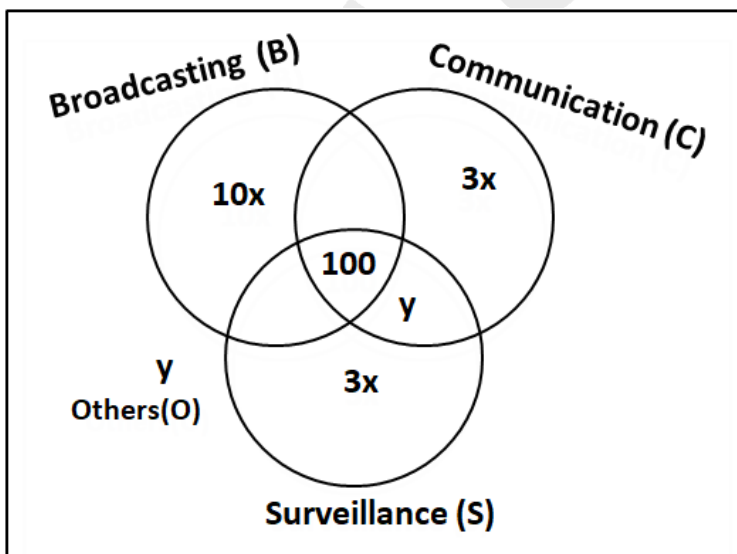
It is given that a satellite serving either B, or C, or S does not serve O. So we can say that it's basically 3 satellites broadcasting (B), communication (C), surveillance (S) which can have intersections. Those satellites which are not part of any category are placed in others. We can draw the Venn diagram as follows.



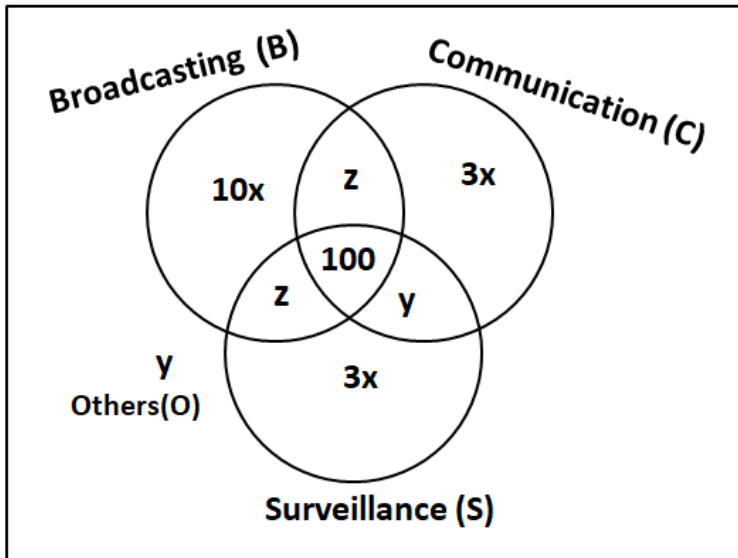
1. The numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1.
 2. The number of satellites serving all three of B, C, and S is 100.
 3. The number of satellites exclusively serving C is the same as the number of satellites exclusively serving S. This number is 30% of the number of satellites exclusively serving B.
 4. The number of satellites serving O is the same as the number of satellites serving both C and S but not B.
- Let '10x' be the number of satellites exclusively serving B. Then, the number of satellites exclusively serving C and S = $0.30 \times 10x = 3x$



Let 'y' be the number of satellites serving others(O).



Let 'z' be the number of satellites serving B, C but not S. Since the numbers of satellites serving B, C, and S (though may be not exclusively) are in the ratio 2:1:1. Therefore, we can say that number of satellites serving B, S but not C = z.



It is given that

$$\Rightarrow 10x + 2z + 2y + 6x + 100 = 1600$$

$$\Rightarrow 8x + z + y = 750 \dots (1)$$

The numbers of satellites serving B, C, and S (though maybe not exclusively) are in the ratio 2:1:1.

$$\Rightarrow \frac{10x + 2z + 100}{z + 100 + 3x + y} = \frac{2}{1}$$

$$\Rightarrow 10x + 2z + 100 = 2(z + 100 + 3x + y)$$

$$\Rightarrow 4x = 100 + 2y$$

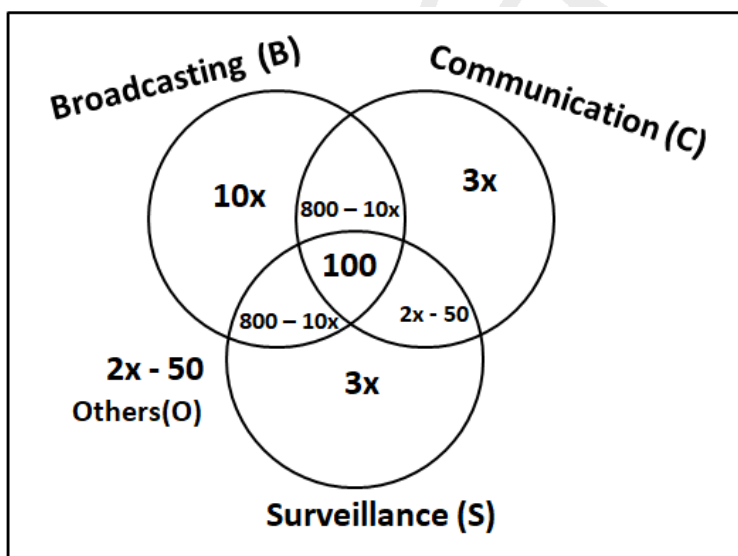
$$\Rightarrow 2x = 50 + y$$

$$\Rightarrow y = 2x - 50 \dots (2)$$

We can substitute this in equation (1)

$$\Rightarrow 8x + z + 2x - 50 = 750$$

$$\Rightarrow z = 800 - 10x \dots (3)$$



Let us define boundary condition for x,

$$\Rightarrow 2x - 50 \geq 0$$

$$\Rightarrow x \geq 25$$

$$\text{Also, } 800 - 10x \geq 0$$

$$\Rightarrow x \leq 80$$

Therefore, we can say that $x \in [25, 80]$.

$$\text{The number of satellites serving C} = 800 - 10x + 100 + 3x + 2x - 50 = 850 - 5x$$

$$\text{At } x = 25, \text{ The number of satellites serving C} = 850 - 5x = 850 - 5 \cdot 25 = 725$$

$$\text{At } x = 80, \text{ The number of satellites serving C} = 850 - 5x = 850 - 5 \cdot 80 = 450$$

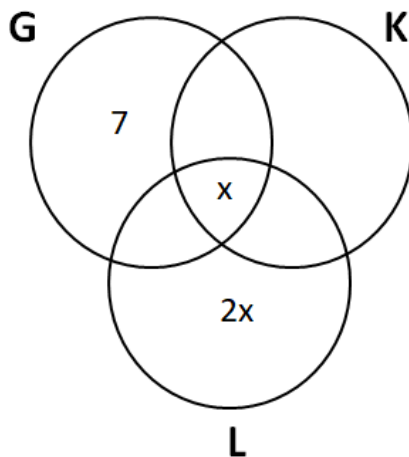
Hence, we can say that the number of satellites serving C must be between 450 and 725. Hence, option C is the correct answer.

 VIDEO SOLUTION

19.2

Let 'x' be the number of students enrolled in all three sports. Then the number of students enrolled only in L = 2x

It is given that there are a total of 17 students enrolled in G. Also, ten students enrolled in G are also enrolled in at least one more sport. Hence, the number of students enrolled in only G = 17 - 10 = 7



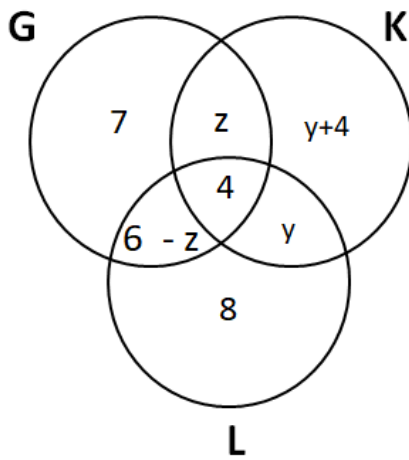
The number of students enrolled only in G is one less than the number of students enrolled only in L. Hence, the number of students enrolled only in L = 7+1

$$\Rightarrow 2x = 8$$

$$\Rightarrow x = 4$$

Let us assume that 'y' students are enrolled in K and L but not G. Then, the number of students enrolled only in K = y + 4

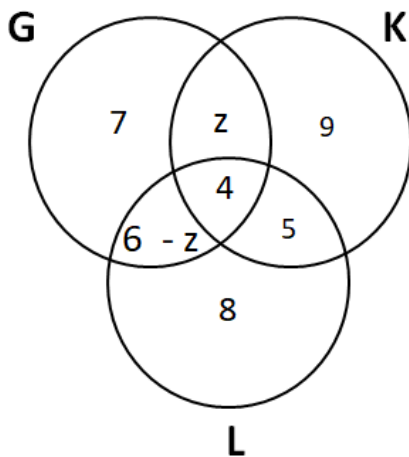
Let us assume that 'z' be the the number of students enrolled in G and K but not L. Then, the number of students enrolled G and L bot not K = 10 - 4 - z = 6 - z



It is given that a total of 39 students in the sports.

$$7 + z + 4 + 6 - z + 8 + y + y + 4 = 39$$

$$\Rightarrow y = 5$$



Number of students enrolled in G = 17

Number of students enrolled in K = $9 + 4 + 5 + z = 18 + z$

Number of students enrolled in L = $6 - z + 4 + 5 + 8 = 23 - z$

It is given that the maximum student enrollment is in L.

$$\Rightarrow 23 - z > 18 + z$$

$$\Rightarrow 2z < 5$$

$$\Rightarrow z < 2.5$$

Therefore, we can say that z can take three values = {0, 1, 2}

Hence, the number of students enrolled in K = $18 + z = \{18, 19, 20\}$

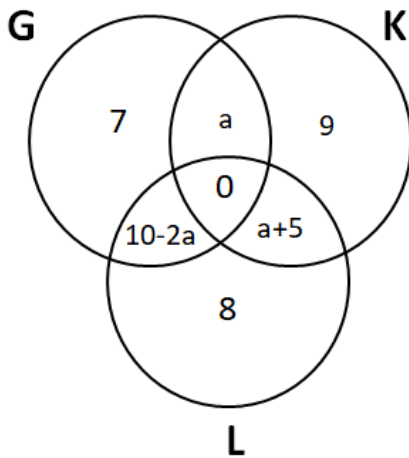
It is given that after withdrawal the number of students enrolled in K went down by one. This one student must have left sports K. Hence we can say that the remaining 3 students must have left either G or L.

Before withdraw there were a total of 24 students were enrolled in exactly 1 sports, 11 students were enrolled in exactly 2 courses and 4 students were enrolled in all three courses.

The students which were enrolled in all three sports, withdrew from one of the sports. Hence, we can say that now the number of students who were enrolled in exactly 2 courses = $11 + 4 = 15$.

It is given that the number of students enrolled in G was six less than the number of students enrolled in L. Let 'a' be the number of students who were enrolled in G and K but not L. Then, the number of students who were enrolled in L and K but not G = $a + 5$

Consequently, we can say that the number of students enrolled in G and L but not K = $15 - (2a + 5) = 10 - 2a$



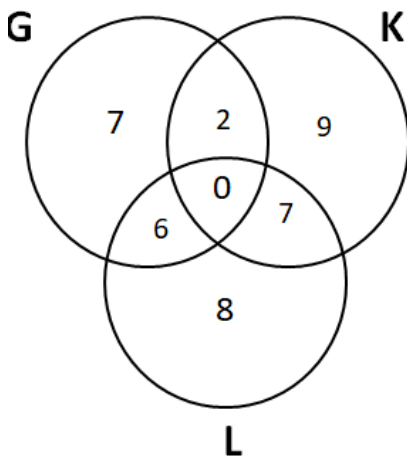
Number of students enrolled in this case = $a + a+5 + 9 = 14 + 2a$. We can see that ' $14+2a$ ' is an even number. It is given that the number of students enrolled in K went down by one. Therefore, we can say that the number of students enrolled in K earlier was an odd number.

Hence, the number of students enrolled in K = $18 + z = \{18, 19, 20\}$

We can see that only '19' is an odd number. Hence, we can say that the number of students enrolled in K after withdrawal = 18

$$\Rightarrow 14 + 2a = 18$$

$$\Rightarrow a = 2$$



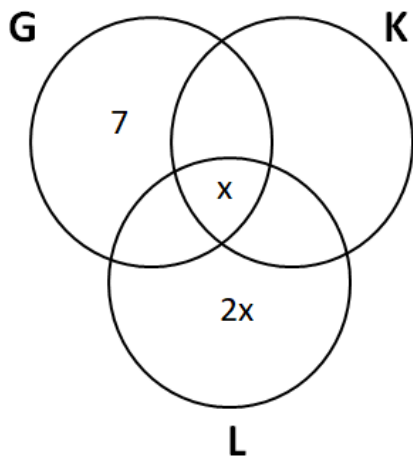
From the diagram we can see that the number of students enrolled in both G and K = 2.

[VIDEO SOLUTION](#)

20. A

Let 'x' be the number of students enrolled in all three sports. Then the number of students enrolled only in L = $2x$

It is given that there are a total of 17 students enrolled in G. Also, ten students enrolled in G are also enrolled in at least one more sport. Hence, the number of students enrolled in only G = $17 - 10 = 7$



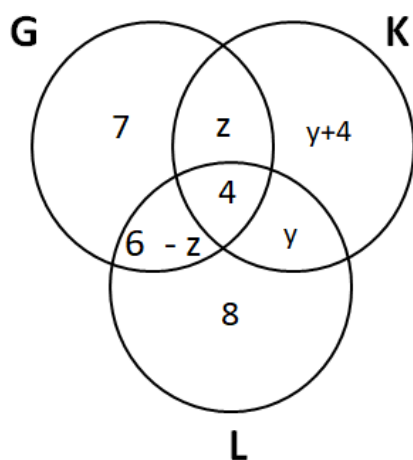
The number of students enrolled only in G is one less than the number of students enrolled only in L. Hence, the number of students enrolled only in L = 7+1

$$\Rightarrow 2x = 8$$

$$\Rightarrow x = 4$$

Let us assume that 'y' students are enrolled in K and L but not G. Then, the number of students enrolled only in K = y + 4

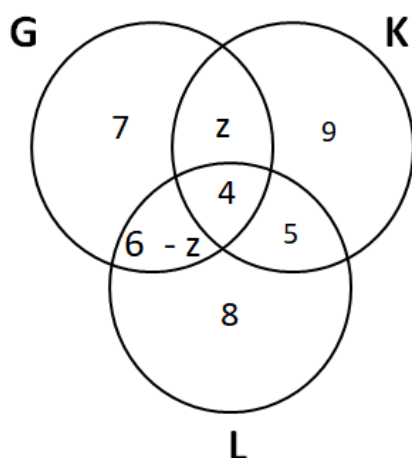
Let us assume that 'z' be the the number of students enrolled in G and K but not L. Then, the number of students enrolled G and L bot not K = 10 - 4 - z = 6 - z



It is given that a total of 39 students in the sports.

$$7 + z + 4 + 6 - z + 8 + y + y + 4 = 39$$

$$\Rightarrow y = 5$$



Number of students enrolled in G = 17

Number of students enrolled in K = $9 + 4 + 5 + z = 18 + z$

Number of students enrolled in L = $6 - z + 4 + 5 + 8 = 23 - z$

It is given that the maximum student enrollment is in L.

$$\Rightarrow 23 - z > 18 + z$$

$$\Rightarrow 2z < 5$$

$$\Rightarrow z < 2.5$$

Therefore, we can say that z can take three values = $\{0, 1, 2\}$

Hence, the number of students enrolled in K = $18 + z = \{18, 19, 20\}$

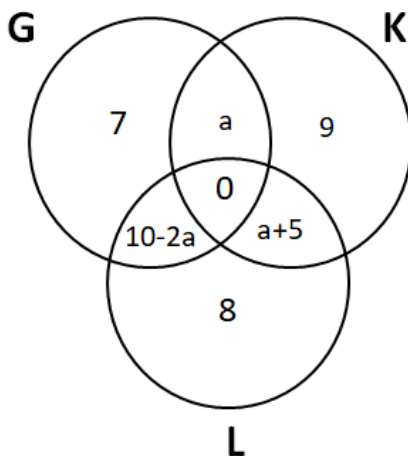
It is given that after withdrawal the number of students enrolled in K went down by one. This one student must have left sports K. Hence we can say that the remaining 3 students must have left either G or L.

Before withdraw there were a total of 24 students were enrolled in exactly 1 sports, 11 students were enrolled in exactly 2 courses and 4 students were enrolled in all three courses.

The students which were enrolled in all three sports, withdrew from one of the sports. Hence, we can say that now the number of students who were enrolled in exactly 2 courses = $11 + 4 = 15$.

It is given that the number of students enrolled in G was six less than the number of students enrolled in L. Let 'a' be the number of students who were enrolled in G and K but not L. Then, the number of students who were enrolled in L and K but not G = $a + 5$

Consequently, we can say that the number of students enrolled in G and L but not K = $15 - (2a + 5) = 10 - 2a$



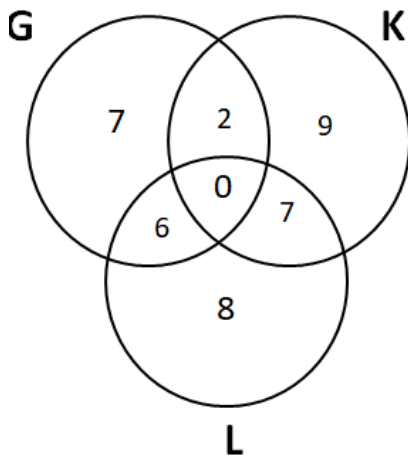
Number of students enrolled in this case = $a + a + 5 + 9 = 14 + 2a$. We can see that '14+2a' is an even number. It is given that the number of students enrolled in K went down by one. Therefore, we can say that the number of students enrolled in K earlier was an odd number.

Hence, the number of students enrolled in K = $18 + z = \{18, 19, 20\}$

We can see that only '19' is an odd number. Hence, we can say that the number of students enrolled in K after withdrawal = 18

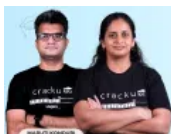
$$\Rightarrow 14 + 2a = 18$$

$$\Rightarrow a = 2$$



From the diagram we can see that the number of students enrolled in both G and L = 6. Hence, option A is the correct answer.

[VIDEO SOLUTION](#)



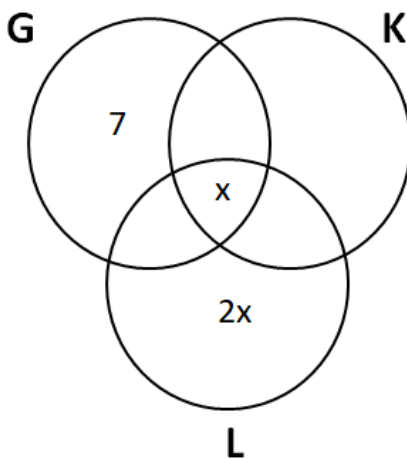
CAT WhatsApp Group



21. C

Let 'x' be the number of students enrolled in all three sports. Then the number of students enrolled only in L = $2x$

It is given that there are a total of 17 students enrolled in G. Also, ten students enrolled in G are also enrolled in at least one more sport. Hence, the number of students enrolled in only G = $17 - 10 = 7$



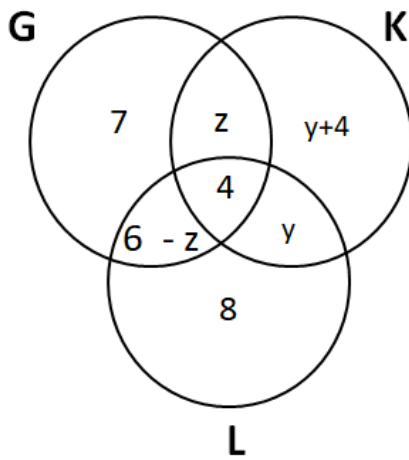
The number of students enrolled only in G is one less than the number of students enrolled only in L. Hence, the number of students enrolled only in L = $7 + 1$

$$\Rightarrow 2x = 8$$

$$\Rightarrow x = 4$$

Let us assume that 'y' students are enrolled in K and L but not G. Then, the number of students enrolled only in K = $y + 4$

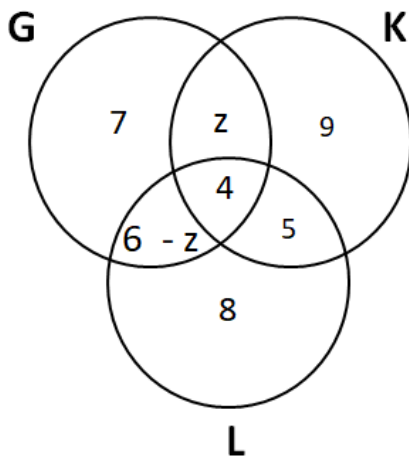
Let us assume that 'z' be the the number of students enrolled in G and K but not L. Then, the number of students enrolled G and L bot not K = $10 - 4 - z = 6 - z$



It is given that a total of 39 students in the sports.

$$7 + z + 4 + 6 - z + 8 + y + y + 4 = 39$$

$$\Rightarrow y = 5$$



Number of students enrolled in G = 17

Number of students enrolled in K = $9 + 4 + 5 + z = 18 + z$

Number of students enrolled in L = $6 - z + 4 + 5 + 8 = 23 - z$

It is given that the maximum student enrollment is in L.

$$\Rightarrow 23 - z > 18 + z$$

$$\Rightarrow 2z < 5$$

$$\Rightarrow z < 2.5$$

Therefore, we can say that z can take three values = {0, 1, 2}

It is given that the numbers of students enrolled in K and L are in the ratio 19:22.

$$\frac{18 + z}{23 - z} = \frac{19}{22}$$

$z = 1$ which is a possible solution as well.

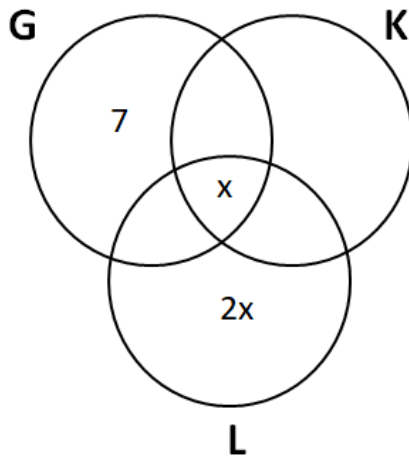
In this case the number of students enrolled in L = $23 - z = 23 - 1 = 22$. Hence, option C is the correct answer.

VIDEO SOLUTION

22.4

Let 'x' be the number of students enrolled in all three sports. Then the number of students enrolled only in L = 2x

It is given that there are a total of 17 students enrolled in G. Also, ten students enrolled in G are also enrolled in at least one more sport. Hence, the number of students enrolled in only G = $17 - 10 = 7$



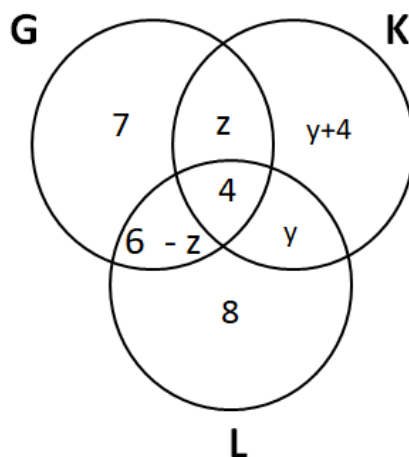
The number of students enrolled only in G is one less than the number of students enrolled only in L. Hence, the number of students enrolled only in L = $7 + 1$

$$\Rightarrow 2x = 8$$

$$\Rightarrow x = 4$$

Let us assume that 'y' students are enrolled in K and L but not G. Then, the number of students enrolled only in K = $y + 4$

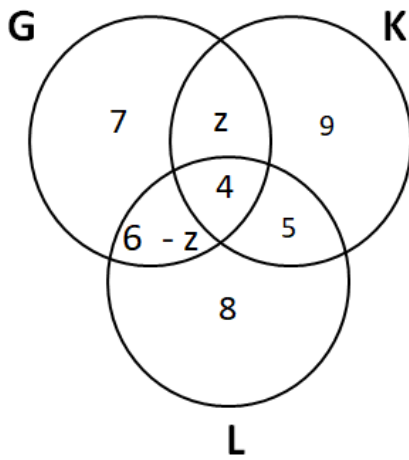
Let us assume that 'z' be the the number of students enrolled in G and K but not L. Then, the number of students enrolled G and L bot not K = $10 - 4 - z = 6 - z$



It is given that a total of 39 students in the sports.

$$7 + z + 4 + 6 - z + 8 + y + y + 4 = 39$$

$$\Rightarrow y = 5$$



Number of students enrolled in G = 17

Number of students enrolled in K = $9 + 4 + 5 + z = 18 + z$

Number of students enrolled in L = $6 - z + 4 + 5 + 8 = 23 - z$

It is given that the maximum student enrollment is in L.

$$\Rightarrow 23 - z > 18 + z$$

$$\Rightarrow 2z < 5$$

$$\Rightarrow z < 2.5$$

Therefore, we can say that z can take three values = $\{0, 1, 2\}$

The number of students enrolled in both G and L but not in K = $6 - z$. This number will be minimum when 'z' is maximum. $z_{\text{max}} = 2$

Therefore, the minimum number of students enrolled in both G and L but not in K = $6 - 2 = 4$

VIDEO SOLUTION

23. **A**

No. of girls = 15

Let the no. of boys be x

No. of singers = 6

No of boys who are singers = 4

Therefore, no of girls who are singers = 2

No of dancers = 10

No of boys who are dancers = 6

Therefore, no. of girls who are dancers = 4

No. of boys who are neither singers nor dancers = $x - 10$

No. of girls who are neither singers nor dancers = 9

	Not interested	1-day event	2-day event	2-day event
Boys (x)				
Singers (4)				
Dancers (6)				
Neither singers nor dancers (x-10)				
Girls (15)				
Singers (2)				
Dancers (4)				
Neither singers nor dancers (9)				

Now we fill the above table,
using statements 1 and 2, we get the following table

	Not interested	1-day event	2-day event	3-day event
Boys (x)	0.2x	0.8x	0.6x	
Singers (4)				
Dancers (6)				
Neither singers nor dancers (x-10)				
Girls (15)	0	15		
Singers (2)	0	2		
Dancers (4)	0	4		
Neither singers nor dancers (9)	0	9		

Let the number of girls who are interested in attending a 2-day event be a and the number of girls who are dancers and are interested in 2-day event be b.

Now using statements 3 and 4, we get

	Not interested	1-day event	2-day event	3-day event
Boys (x)	0.2x	0.8x	0.6x	
Singers (4)	0	4	4	4
Dancers (6)			0.18x-4	2
Neither singers nor dancers (x-10)			0.42x	
Girls (15)	0	15	a	0
Singers (2)	0	2	0.4a-b	0
Dancers (4)	0	4	b	0
Neither singers nor dancers (9)	0	9	0.6a	0

$$2 \leq 0.18x - 4 \leq 6$$

$$6 \leq 0.18x \leq 10$$

0.18x should be integer for which x should be a multiple of 50, and 0.18x lies between 6 and 10; therefore, the only possible value of x is 50.

	Not interested	1-day event	2-day event	3-day event
Boys (50)	10	40	30	
Singers (4)	0	4	4	4
Dancers (6)			5	2
Neither singers nor dancers (40)			21	
Girls (15)	0	15	a	0
Singers (2)	0	2	0.4a-b	0
Dancers (4)	0	4	b	0
Neither singers nor dancers (9)	0	9	0.6a	0

From statement 5, we can say that,

$$4 + 0.4a - b = 5 + b + 1$$

$$\text{or, } 0.4a = 2 + 2b$$

$$\text{or, } a = 5(1+b)$$

a should be a multiple of 5 as b is a whole number. So possible values of a can be 5, 10 or 15. Now, as the maximum value of b can be 4 and the maximum value of 0.4a-b can be 2, so the only possible value of a satisfying the conditions above is 5. If a= 5 then b=1.

	Not interested	1-day event	2-day event	3-day event
Boys (50)	10	40	30	
Singers (4)	0	4	4	4
Dancers (6)	0/1	6/5	5	2
Neither singers nor dancers (40)	10/9	30/31	21	
Girls (15)	0	15	5	0
Singers (2)	0	2	2	0
Dancers (4)	0	4	0	0
Neither singers nor dancers (9)	0	9	3	0

VIDEO SOLUTION

24.50

No. of girls = 15

Let the no. of boys be x

No. of singers = 6

No of boys who are singers = 4

Therefore, no of girls who are singers = 2

No of dancers = 10

No of boys who are dancers = 6

Therefore, no. of girls who are dancers = 4

No. of boys who are neither singers nor dancers = $x-10$

No. of girls who are neither singers nor dancers = 9

	Not interested	1-day event	2-day event	2-day event
Boys (x)				
Singers (4)				
Dancers (6)				
Neither singers nor dancers (x-10)				
Girls (15)				
Singers (2)				
Dancers (4)				
Neither singers nor dancers (9)				

Now we fill the above table,
using statements 1 and 2, we get the following table

	Not interested	1-day event	2-day event	3-day event
Boys (x)	0.2x	0.8x	0.6x	
Singers (4)				
Dancers (6)				
Neither singers nor dancers (x-10)				
Girls (15)	0	15		
Singers (2)	0	2		
Dancers (4)	0	4		
Neither singers nor dancers (9)	0	9		

Let the number of girls who are interested in attending a 2-day event be a and the number of girls who are dancers and are interested in 2-day event be b .

Now using statements 3 and 4, we get

	Not interested	1-day event	2-day event	3-day event
Boys (x)	0.2x	0.8x	0.6x	
Singers (4)	0	4	4	4
Dancers (6)			$0.18x - 4$	2
Neither singers nor dancers (x-10)			$0.42x$	
Girls (15)	0	15	a	0
Singers (2)	0	2	$0.4a - b$	0
Dancers (4)	0	4	b	0
Neither singers nor dancers (9)	0	9	$0.6a$	0

$$2 \leq 0.18x - 4 \leq 6$$

$$6 \leq 0.18x \leq 10$$

$0.18x$ should be integer for which x should be a multiple of 50, and $0.18x$ lies between 6 and 10; therefore, the only possible value of x is 50.

	Not interested	1-day event	2-day event	3-day event
Boys (50)	10	40	30	
Singers (4)	0	4	4	4
Dancers (6)			5	2
Neither singers nor dancers (40)			21	
Girls (15)	0	15	a	0
Singers (2)	0	2	$0.4a - b$	0
Dancers (4)	0	4	b	0
Neither singers nor dancers (9)	0	9	$0.6a$	0

From statement 5, we can say that,

$$4 + 0.4a - b = 5 + b + 1$$

$$\text{or, } 0.4a = 2 + 2b$$

$$\text{or, } a = 5(1 + b)$$

a should be a multiple of 5 as b is a whole number. So possible values of a can be 5, 10 or 15. Now, as the maximum value of b can be 4 and the maximum value of $0.4a - b$ can be 2, so the only possible value of a satisfying the conditions above is 5. If $a = 5$ then $b = 1$.

	Not interested	1-day event	2-day event	3-day event
Boys (50)	10	40	30	
Singers (4)	0	4	4	4
Dancers (6)	0/1	6/5	5	2
Neither singers nor dancers (40)	10/9	30/31	21	
Girls (15)	0	15	5	0
Singers (2)	0	2	2	0
Dancers (4)	0	4	0	0
Neither singers nor dancers (9)	0	9	3	0

VIDEO SOLUTION

25. C

No. of girls = 15

Let the no. of boys be x

No. of singers = 6

No of boys who are singers = 4

Therefore, no of girls who are singers = 2

No of dancers = 10

No of boys who are dancers = 6

Therefore, no. of girls who are dancers = 4

No. of boys who are neither singers nor dancers = $x-10$

No. of girls who are neither singers nor dancers = 9

	Not interested	1-day event	2-day event	2-day event
Boys (x)				
Singers (4)				
Dancers (6)				
Neither singers nor dancers (x-10)				
Girls (15)				
Singers (2)				
Dancers (4)				
Neither singers nor dancers (9)				

Now we fill the above table,
using statements 1 and 2, we get the following table

	Not interested	1-day event	2-day event	3-day event
Boys (x)	0.2x	0.8x	0.6x	
Singers (4)				
Dancers (6)				
Neither singers nor dancers (x-10)				
Girls (15)	0	15		
Singers (2)	0	2		
Dancers (4)	0	4		
Neither singers nor dancers (9)	0	9		

Let the number of girls who are interested in attending a 2-day event be a and the number of girls who are dancers and are interested in 2-day event be b.

Now using statements 3 and 4, we get

	Not interested	1-day event	2-day event	3-day event
Boys (x)	0.2x	0.8x	0.6x	
Singers (4)	0	4	4	4
Dancers (6)			$0.18x - 4$	2
Neither singers nor dancers (x-10)			$0.42x$	
Girls (15)	0	15	a	0
Singers (2)	0	2	$0.4a - b$	0
Dancers (4)	0	4	b	0
Neither singers nor dancers (9)	0	9	$0.6a$	0

$$2 \leq 0.18x - 4 \leq 6$$

$$6 \leq 0.18x \leq 10$$

$0.18x$ should be integer for which x should be a multiple of 50, and $0.18x$ lies between 6 and 10; therefore, the only possible value of x is 50.

	Not interested	1-day event	2-day event	3-day event
Boys (50)	10	40	30	
Singers (4)	0	4	4	4
Dancers (6)			5	2
Neither singers nor dancers (40)			21	
Girls (15)	0	15	a	0
Singers (2)	0	2	$0.4a - b$	0
Dancers (4)	0	4	b	0
Neither singers nor dancers (9)	0	9	$0.6a$	0

From statement 5, we can say that,

$$4 + 0.4a - b = 5 + b + 1$$

$$\text{or, } 0.4a = 2 + 2b$$

$$\text{or, } a = 5(1 + b)$$

a should be a multiple of 5 as b is a whole number. So possible values of a can be 5, 10 or 15. Now, as the maximum value of b can be 4 and the maximum value of $0.4a - b$ can be 2, so the only possible value of a satisfying the conditions above is 5. If $a = 5$ then $b = 1$.

	Not interested	1-day event	2-day event	3-day event
Boys (50)	10	40	30	
Singers (4)	0	4	4	4
Dancers (6)	0/1	6/5	5	2
Neither singers nor dancers (40)	10/9	30/31	21	
Girls (15)	0	15	5	0
Singers (2)	0	2	2	0
Dancers (4)	0	4	0	0
Neither singers nor dancers (9)	0	9	3	0

VIDEO SOLUTION



CAT Mock Test

Mocks designed exactly like the actual CAT



26. C

No. of girls = 15

Let the no. of boys be x

No. of singers = 6

No of boys who are singers = 4

Therefore, no of girls who are singers = 2

No of dancers = 10

No of boys who are dancers = 6

Therefore, no. of girls who are dancers = 4

No. of boys who are neither singers nor dancers = $x-10$

No. of girls who are neither singers nor dancers = 9

	Not interested	1-day event	2-day event	2-day event
Boys (x)				
Singers (4)				
Dancers (6)				
Neither singers nor dancers (x-10)				
Girls (15)				
Singers (2)				
Dancers (4)				
Neither singers nor dancers (9)				

Now we fill the above table,
using statements 1 and 2, we get the following table

	Not interested	1-day event	2-day event	3-day event
Boys (x)	0.2x	0.8x	0.6x	
Singers (4)				
Dancers (6)				
Neither singers nor dancers (x-10)				
Girls (15)	0	15		
Singers (2)	0	2		
Dancers (4)	0	4		
Neither singers nor dancers (9)	0	9		

Let the number of girls who are interested in attending a 2-day event be a and the number of girls who are dancers and are interested in 2-day event be b .

Now using statements 3 and 4, we get

	Not interested	1-day event	2-day event	3-day event
Boys (50)	10	40	30	
Singers (4)	0	4	4	4
Dancers (6)	0/1	6/5	5	2
Neither singers nor dancers (40)	10/9	30/31	21	
Girls (15)	0	15	5	0
Singers (2)	0	2	2	0
Dancers (4)	0	4	0	0
Neither singers nor dancers (9)	0	9	3	0

VIDEO SOLUTION

27. B

No. of girls = 15

Let the no. of boys be x

No. of singers = 6

No of boys who are singers = 4

Therefore, no of girls who are singers = 2

No of dancers = 10

No of boys who are dancers = 6

Therefore, no. of girls who are dancers = 4

No. of boys who are neither singers nor dancers = $x-10$

No. of girls who are neither singers nor dancers = 9

	Not interested	1-day event	2-day event	2-day event
Boys (x)				
Singers (4)				
Dancers (6)				
Neither singers nor dancers (x-10)				
Girls (15)				
Singers (2)				
Dancers (4)				
Neither singers nor dancers (9)				

Now we fill the above table,
using statements 1 and 2, we get the following table

	Not interested	1-day event	2-day event	3-day event
Boys (x)	0.2x	0.8x	0.6x	
Singers (4)				
Dancers (6)				
Neither singers nor dancers (x-10)				
Girls (15)	0	15		
Singers (2)	0	2		
Dancers (4)	0	4		
Neither singers nor dancers (9)	0	9		

Let the number of girls who are interested in attending a 2-day event be a and the number of girls who are dancers and are interested in 2-day event be b.

Now using statements 3 and 4, we get

	Not interested	1-day event	2-day event	3-day event
Boys (50)	10	40	30	
Singers (4)	0	4	4	4
Dancers (6)	0/1	6/5	5	2
Neither singers nor dancers (40)	10/9	30/31	21	
Girls (15)	0	15	5	0
Singers (2)	0	2	2	0
Dancers (4)	0	4	0	0
Neither singers nor dancers (9)	0	9	3	0

Total students interested in 2-day event = 30+5 = 35

Total students = 15 + 50 = 65

Fraction = $\frac{35}{65} = \frac{7}{13}$

VIDEO SOLUTION

28. A

Total number of students surveyed= 500

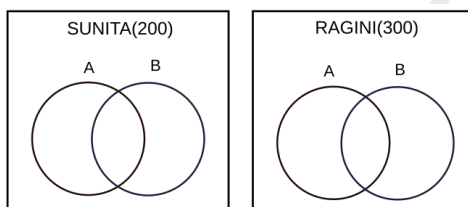
Every student prefers one of the two candidates. Ragini(R) and Sunita(S).

Thus, R+S=500.

According to statement 2, "Among the 200 students who preferred Sunita as student union president, 80% supported proposal A."

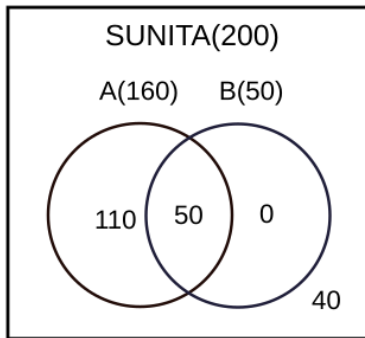
The number of students who support Sunita(S)=200

The number of students who supported Ragini(R)=300

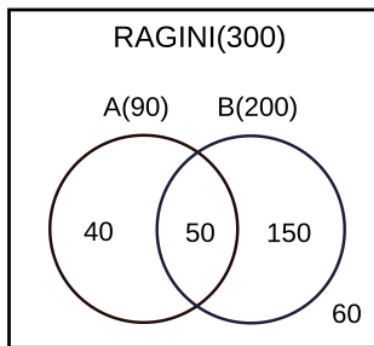


According to statements 2 and 3, 160 students who supported Sunita also supported the proposal A & 90 students who supported Ragini also supported proposal A.

According to statements 4 and 6, we can make the following Venn diagram for Sunita.



According to statement 5 and 7, we can make the following Venn diagram.



From the diagram, we can understand that option A is correct.

[VIDEO SOLUTION](#)

29. D

Total number of students surveyed = 500

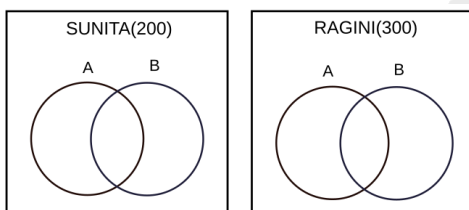
Every student prefers one of the two candidates. Ragini(R) and Sunita(S).

Thus, $R + S = 500$.

According to statement 2, "Among the 200 students who preferred Sunita as student union president, 80% supported proposal A."

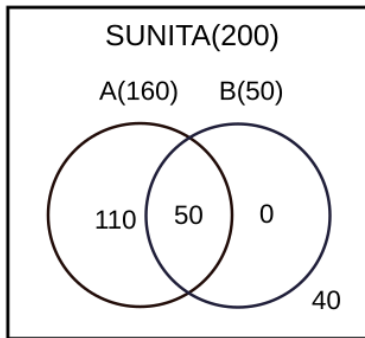
The number of students who support Sunita(S) = 200

The number of students who supported Ragini(R) = 300

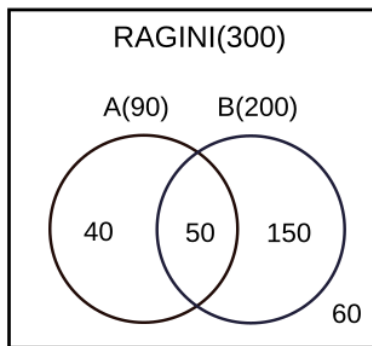


According to statements 2 and 3, 160 students who supported Sunita also supported the proposal A & 90 students who supported Ragini also supported proposal A.

According to statements 4 and 6, we can make the following Venn diagram for Sunita.



According to statement 5 and 7, we can make the following Venn diagram.



According to the Venn diagram, the students surveyed who supported both proposals A and B preferred Sunita as student union president $\frac{50}{50+50} \% = 50\%$

[VIDEO SOLUTION](#)

30.84

Total number of students surveyed = 500

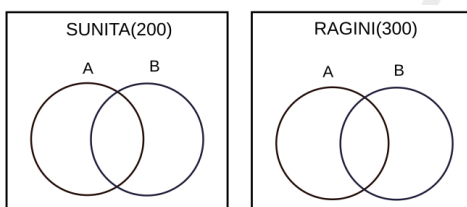
Every student prefers one of the two candidates. Ragini(R) and Sunita(S).

Thus, $R+S=500$.

According to statement 2, "Among the 200 students who preferred Sunita as student union president, 80% supported proposal A."

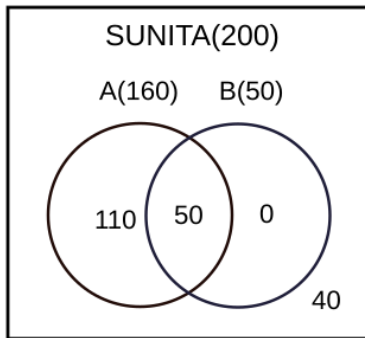
The number of students who support Sunita(S)=200

The number of students who supported Ragini(R)=300

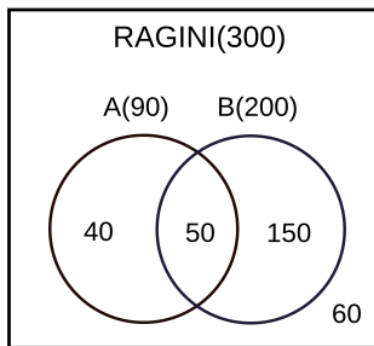


According to statements 2 and 3, 160 students who supported Sunita also supported the proposal A & 90 students who supported Ragini also supported proposal A.

According to statements 4 and 6, we can make the following Venn diagram for Sunita.



According to statement 5 and 7, we can make the following Venn diagram.



The percentage of the students surveyed who did not support proposal A preferred Ragini as student union president = $210/250 = 84\%$

Answer 84

[VIDEO SOLUTION](#)



CAT Online Coaching

your road to IIM stars here



31.64

Total number of students surveyed = 500

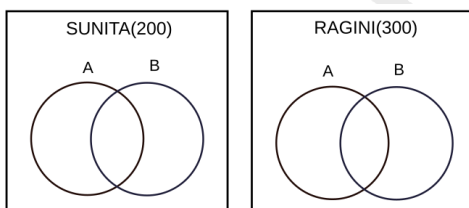
Every student prefers one of the two candidates. Ragini(R) and Sunita(S).

Thus, $R + S = 500$.

According to statement 2, "Among the 200 students who preferred Sunita as student union president, 80% supported proposal A."

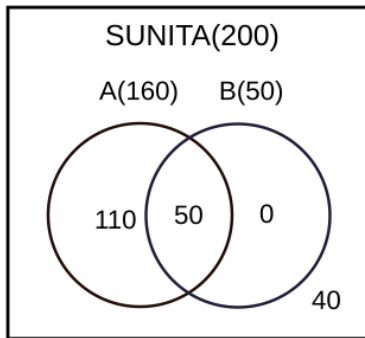
The number of students who support Sunita(S) = 200

The number of students who supported Ragini(R) = 300

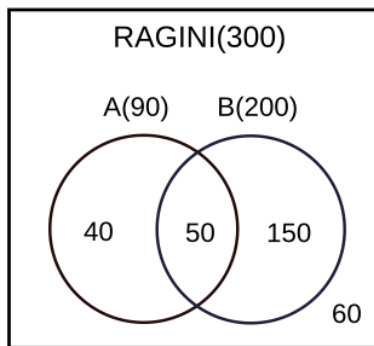


According to statements 2 and 3, 160 students who supported Sunita also supported the proposal A & 90 students who supported Ragini also supported proposal A.

According to statements 4 and 6, we can make the following Venn diagram for Sunita.



According to statement 5 and 7, we can make the following Venn diagram.



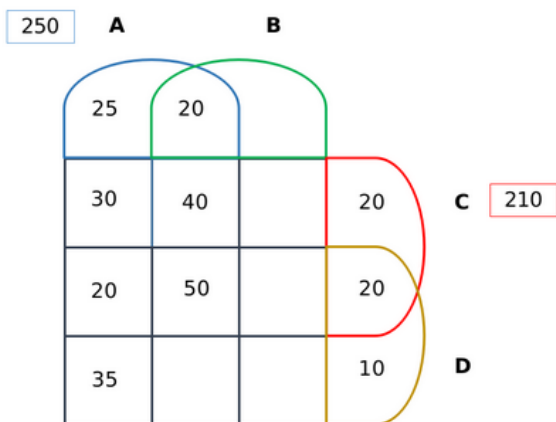
The number of students who preferred Sunita and the proposal A=160

$$=160/250= 64\%$$

[VIDEO SOLUTION](#)

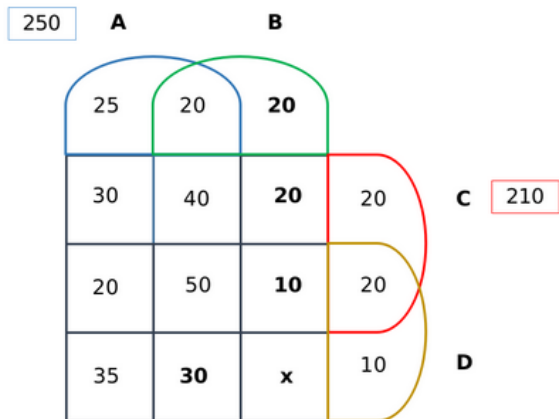
32. 325

Of the 1000 subjects, only 500 have been considered for the treatment. This constitutes our sample set. Thus the four drugs- A, B, C and D have been administered to this set of 500 individuals, while the rest 500 have been given the placebo. Based on the given information, we can then draw the following 4-set Venn diagram:

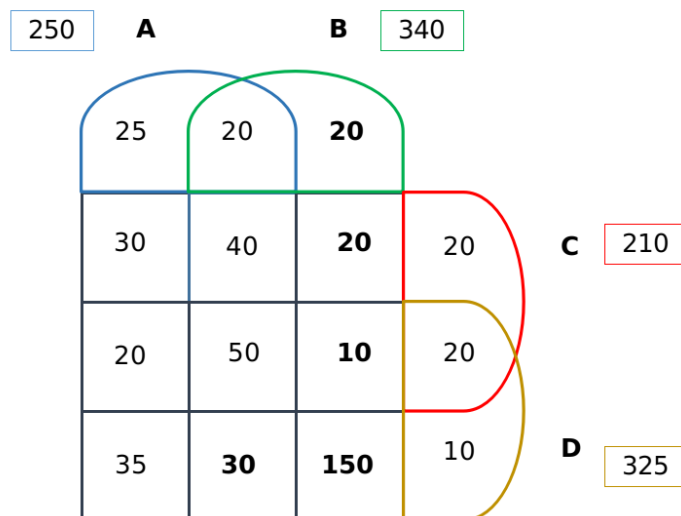


We can solve for the number of patients who were administered the drugs A, B and D excluding C by putting in the values for set A. The required value = $250 - (25+20+30+40+20+50+35) = 30$. Based on condition (c), we know that 100 patients were treated with exactly three types of medicines. Thus, we can fill the slot for the number of patients who were administered only B, C and D excluding A by $100 - (40+20+30) = 10$.

Similarly, based on condition (f), we know that the candidates who were administered only drug B are $75 - (25+20+10) = 20$. Post this, we can easily calculate the number of people administered with only drugs B and C by $210 - (30+20+40+50+10+20+20) = 20$. We can fill in the above values to obtain the following diagram:



The sum of all the values should add up to 500. On solving for 'x' [which represents the number of people who were administered drugs B and D only], we obtain $x = 150$. The final representation would appear as follows:



The number of patients who were treated with medicine type D was **325**.

[VIDEO SOLUTION](#)

33. 340

Of the 1000 subjects, only 500 have been considered for the treatment. This constitutes our sample set. Thus the four drugs- A, B, C and D have been administered to this set of 500 individuals, while the rest 500 have been given the placebo. Based on the given information, we can then draw the following 4-set Venn diagram:

250	A	B		
	25	20		
	30	40		20
	20	50		20
	35			10
				D

C 210

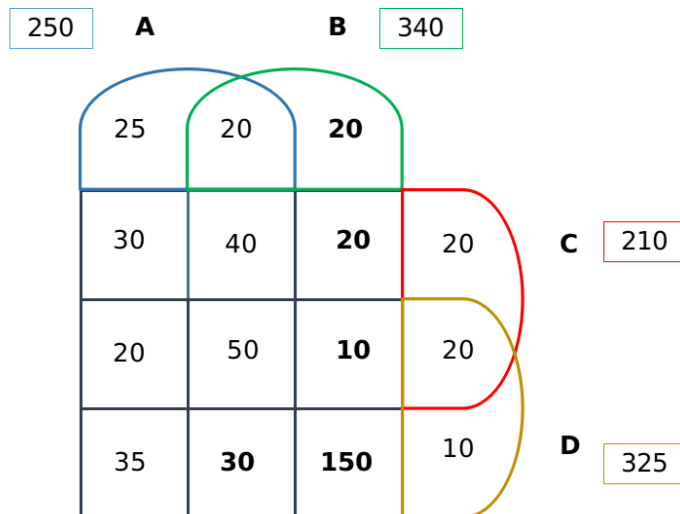
We can solve for the number of patients who were administered the drugs A, B and D excluding C by putting in the values for set A. The required value = $250 - (25+20+30+40+20+50+35) = 30$. Based on condition (c), we know that 100 patients were treated with exactly three types of medicines. Thus, we can fill the slot for the number of patients who were administered only B, C and D excluding A by $100 - (40+20+30) = 10$.

Similarly, based on condition (f), we know that the candidates who were administered only drug B are $75 - (25+20+10) = 20$. Post this, we can easily calculate the number of people administered with only drugs B and C by $210 - (30+20+40+50+10+20+20) = 20$. We can fill in the above values to obtain the following diagram:

250	A	B		
	25	20	20	
	30	40	20	20
	20	50	10	20
	35	30	x	10
				D

C 210

The sum of all the values should add up to 500. On solving for 'x' [which represents the number of people who were administered drugs B and D only], we obtain $x = 150$. The final representation would appear as follows:

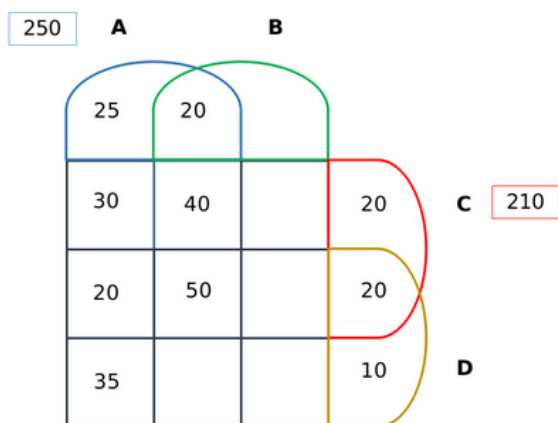


Based on the above, the number of patients who were treated with medicine type B is equal to **340**.

[VIDEO SOLUTION](#)

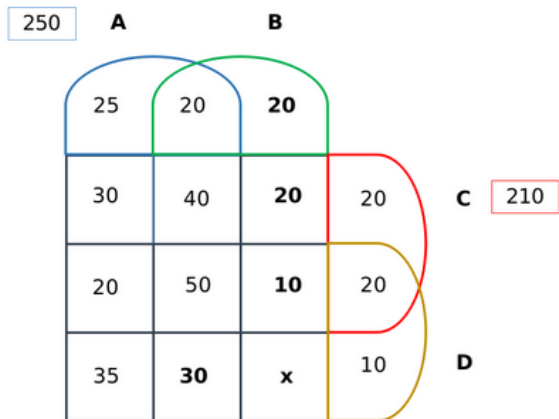
34.10

Of the 1000 subjects, only 500 have been considered for the treatment. This constitutes our sample set. Thus the four drugs- A, B, C and D have been administered to this set of 500 individuals, while the rest 500 have been given the placebo. Based on the given information, we can then draw the following 4-set Venn diagram:

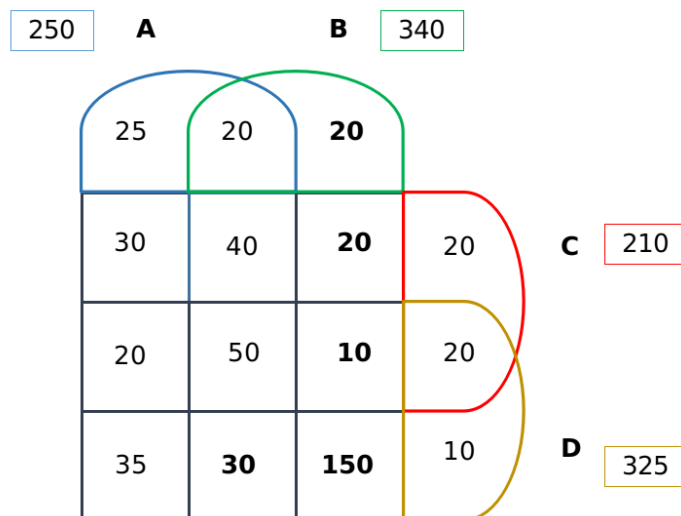


We can solve for the number of patients who were administered the drugs A, B and D excluding C by putting in the values for set A. The required value = $250 - (25 + 20 + 30 + 40 + 20 + 50 + 35) = 30$. Based on condition (c), we know that 100 patients were treated with exactly three types of medicines. Thus, we can fill the slot for the number of patients who were administered only B, C and D excluding A by $100 - (40 + 20 + 30) = 10$.

Similarly, based on condition (f), we know that the candidates who were administered only drug B are $75 - (25 + 20 + 10) = 20$. Post this, we can easily calculate the number of people administered with only drugs B and C by $210 - (30 + 20 + 40 + 50 + 10 + 20 + 20) = 20$. We can fill in the above values to obtain the following diagram:



The sum of all the values should add up to 500. On solving for 'x' [which represents the number of people who were administered drugs B and D only], we obtain $x = 150$. The final representation would appear as follows:



The number of patients who were treated with medicine types B, C and D, but not type A was: **10**.

[VIDEO SOLUTION](#)

35. **150**

Of the 1000 subjects, only 500 have been considered for the treatment. This constitutes our sample set. Thus the four drugs- A, B, C and D have been administered to this set of 500 individuals, while the rest 500 have been given the placebo. Based on the given information, we can then draw the following 4-set Venn diagram:

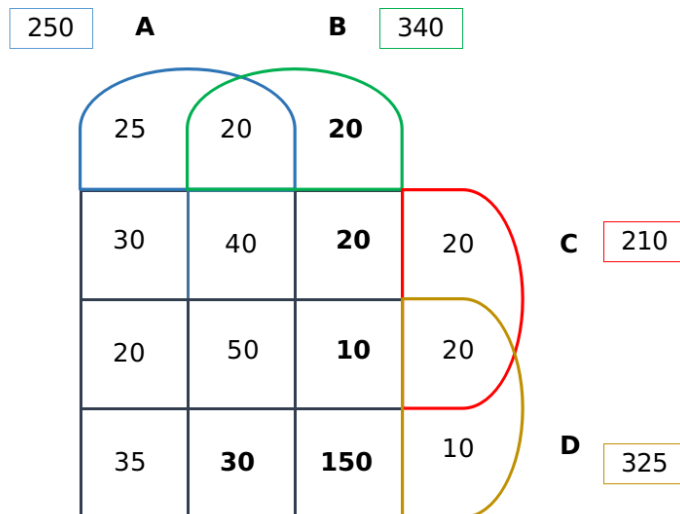
250	A	B		
25	20			
30	40		20	C 210
20	50		20	
35			10	D

We can solve for the number of patients who were administered the drugs A, B and D excluding C by putting in the values for set A. The required value = $250 - (25+20+30+40+20+50+35) = 30$. Based on condition (c), we know that 100 patients were treated with exactly three types of medicines. Thus, we can fill the slot for the number of patients who were administered only B, C and D excluding A by $100 - (40+20+30) = 10$.

Similarly, based on condition (f), we know that the candidates who were administered only drug B are $75 - (25+20+10) = 20$. Post this, we can easily calculate the number of people administered with only drugs B and C by $210 - (30+20+40+50+10+20+20) = 20$. We can fill in the above values to obtain the following diagram:

250	A	B		
25	20	20		
30	40	20	20	C 210
20	50	10	20	
35	30	x	10	D

The sum of all the values should add up to 500. On solving for 'x' [which represents the number of people who were administered drugs B and D only], we obtain $x = 150$. The final representation would appear as follows:



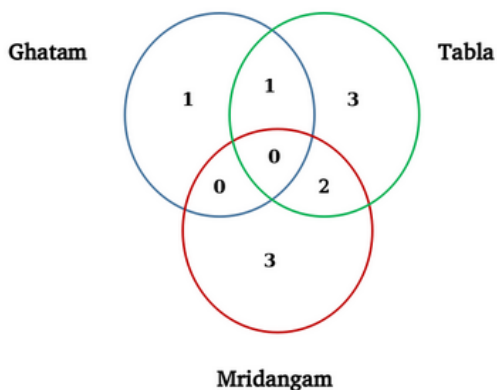
The number of people who were administered drugs B and D only were **150**.

[VIDEO SOLUTION](#)



36. D

Based on the given information, we can form the following Venn-diagram for ease of understanding:



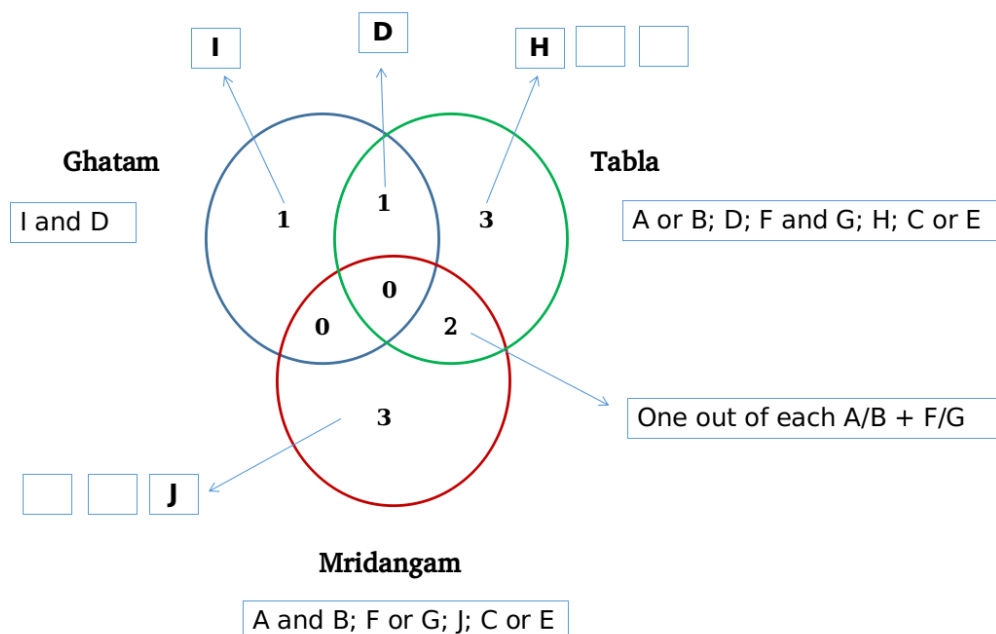
The conditions help us to further bifurcate the individuals based on their expertise.

Mridangam: A and B (condition 1); one out of F and G (condition 3)

Tabla: one out of A and B (condition 1); F and G (condition 3); D (condition 2)

Ghatam: D (condition 2)

Based on condition 4, we infer that 'I' and 'J' are either experts in Ghatam or Mridangam. However, condition 5 adds that 'I' is not an expert in Mridangam. This helps us definitively zero-in on 'I' as an expert in Ghatam. Since 'I' is a Ghatam expert, J is an expert in Mridangam and 'H' is an expert in Tabla [based on conditions 4 and 5]. Thus, we can depict our understanding so far as follows:



Mridangam [total: 5] - A and B (condition 1); one out of F and G (condition 3); J (condition 4 and 5); one out of C and E {remaining experts}

Tabla [total: 6] - one out of A and B (condition 1); F and G (condition 3); D (condition 2); H (condition 4 and 5); one out of C and E {remaining experts}

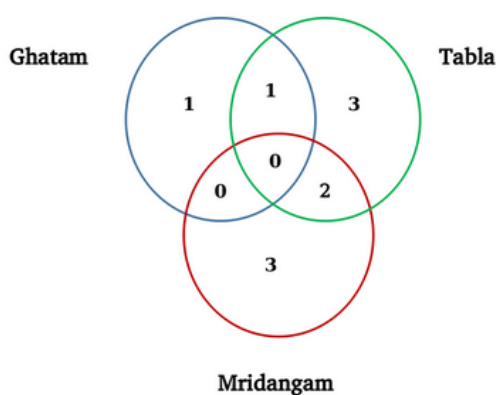
Ghatam [total: 2] - D (condition 2); I (condition 4 and 5)

Thus, we observe that **H** is definitely an expert in tabla but not in either mridangam or ghatam. Hence, Option D is the correct answer.

[VIDEO SOLUTION](#)

37. B

Based on the given information, we can form the following Venn-diagram for ease of understanding:



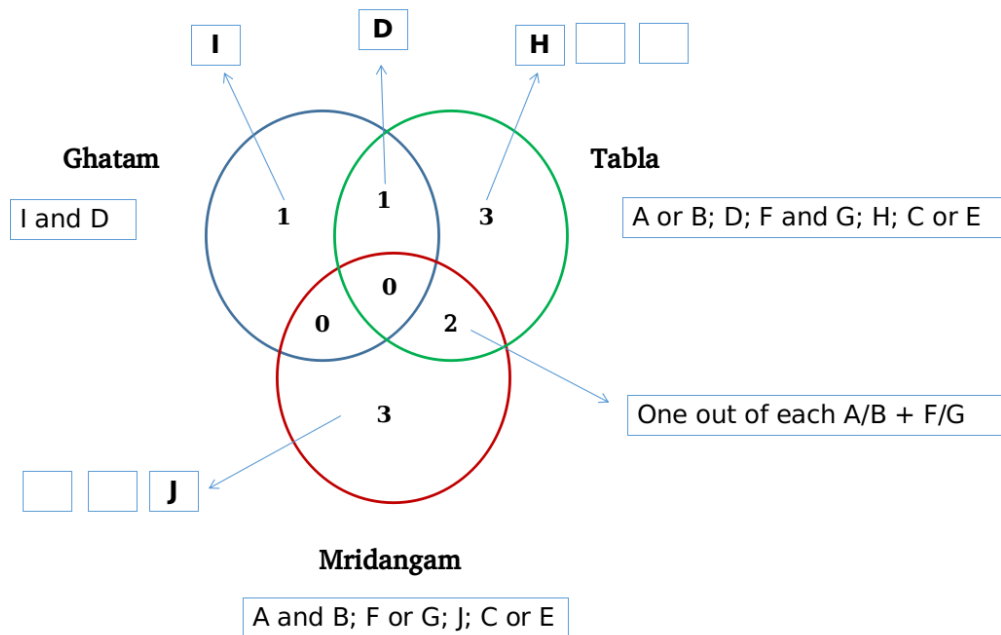
The conditions help us to further bifurcate the individuals based on their expertise.

Mridangam: A and B (condition 1); one out of F and G (condition 3)

Tabla: one out of A and B (condition 1); F and G (condition 3); D (condition 2)

Ghatam: D (condition 2)

Based on condition 4, we infer that 'I' and 'J' are either experts in Ghatam or Mridangam. However, condition 5 adds that 'I' is not an expert in Mridangam. This helps us definitively zero-in on 'I' as an expert in Ghatam. Since 'I' is a Ghatam expert, J is an expert in Mridangam and 'H' is an expert in Tabla [based on conditions 4 and 5]. Thus, we can depict our understanding so far as follows:



Mridangam [total: 5] - A and B (condition 1); one out of F and G (condition 3); J (condition 4 and 5); one out of C and E {remaining experts}

Tabla [total: 6] - one out of A and B (condition 1); F and G (condition 3); D (condition 2); H (condition 4 and 5); one out of C and E {remaining experts}

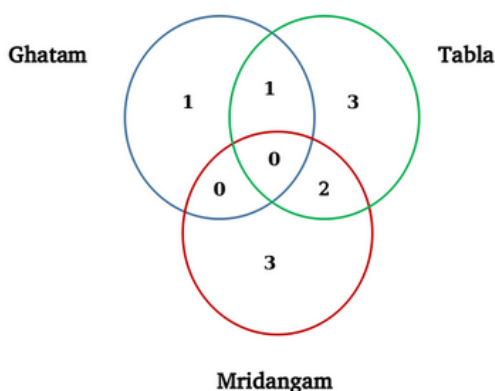
Ghatam [total: 2] - D (condition 2); I (condition 4 and 5)

We observe that the pair **C and E** cannot have any musician who is an expert in both tabla and mridangam but not in ghatam. Hence, Option B is the correct answer.

VIDEO SOLUTION

38. A

Based on the given information, we can form the following Venn-diagram for ease of understanding:



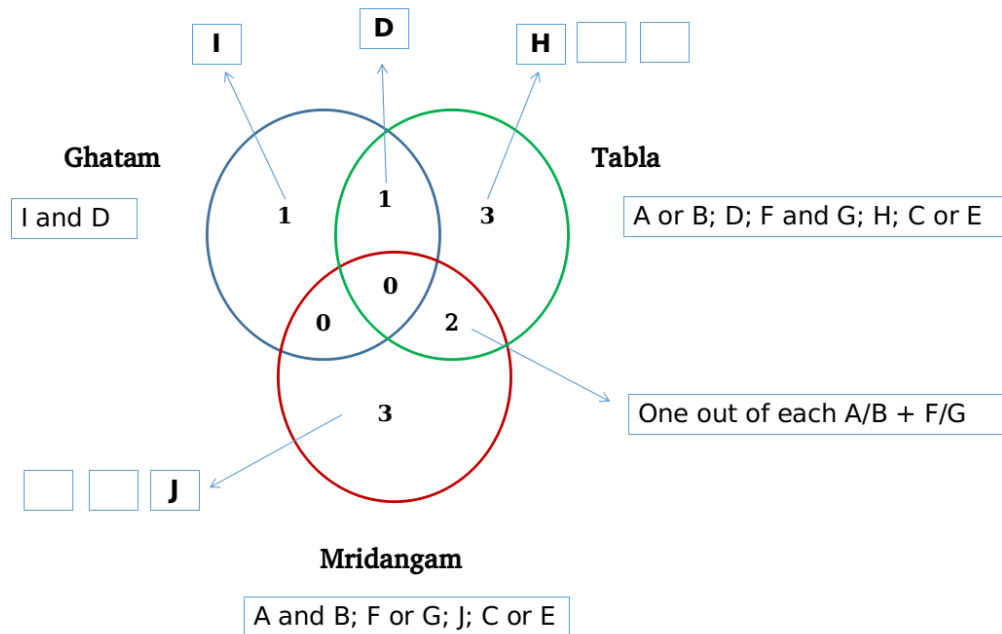
The conditions help us to further bifurcate the individuals based on their expertise.

Mridangam: A and B (condition 1); one out of F and G (condition 3)

Tabla: one out of A and B (condition 1); F and G (condition 3); D (condition 2)

Ghatam: D (condition 2)

Based on condition 4, we infer that 'I' and 'J' are either experts in Ghatam or Mridangam. However, condition 5 adds that 'I' is not an expert in Mridangam. This helps us definitively zero-in on 'I' as an expert in Ghatam. Since 'I' is a Ghatam expert, J is an expert in Mridangam and 'H' is an expert in Tabla [based on conditions 4 and 5]. Thus, we can depict our understanding so far as follows:



Mridangam [total: 5] - A and B (condition 1); one out of F and G (condition 3); J (condition 4 and 5); one out of C and E {remaining experts}

Tabla [total: 6] - one out of A and B (condition 1); F and G (condition 3); D (condition 2); H (condition 4 and 5); one out of C and E {remaining experts}

Ghatam [total: 2] - D (condition 2); I (condition 4 and 5)

If C is an expert in Mridangam, E has to be an expert in Tabla. Additionally, if F is not an expert in Mridangam, he has to be an expert only in Tabla while G will be an expert in both Tabla and Mridangam.

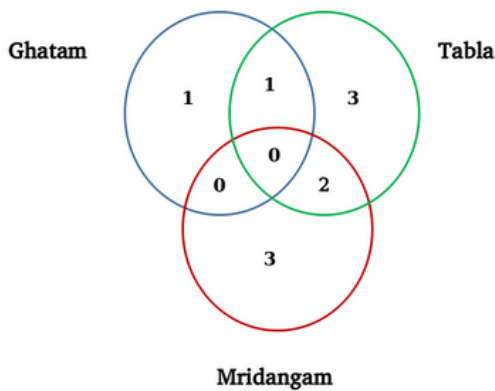
Tabla [total: 6] - one out of A and B (condition 1); F and G (condition 3); D (condition 2); H (condition 4 and 5); E {based on the condition given in the question}

Of these experts, A, B and G are experts of Mridangam and D is an expert of Ghatam as well. Thus, excluding these, we have F, H and E who are the experts solely in Tabla. Thus, Option A is the correct answer.

VIDEO SOLUTION

39. B

Based on the given information, we can form the following Venn-diagram for ease of understanding:



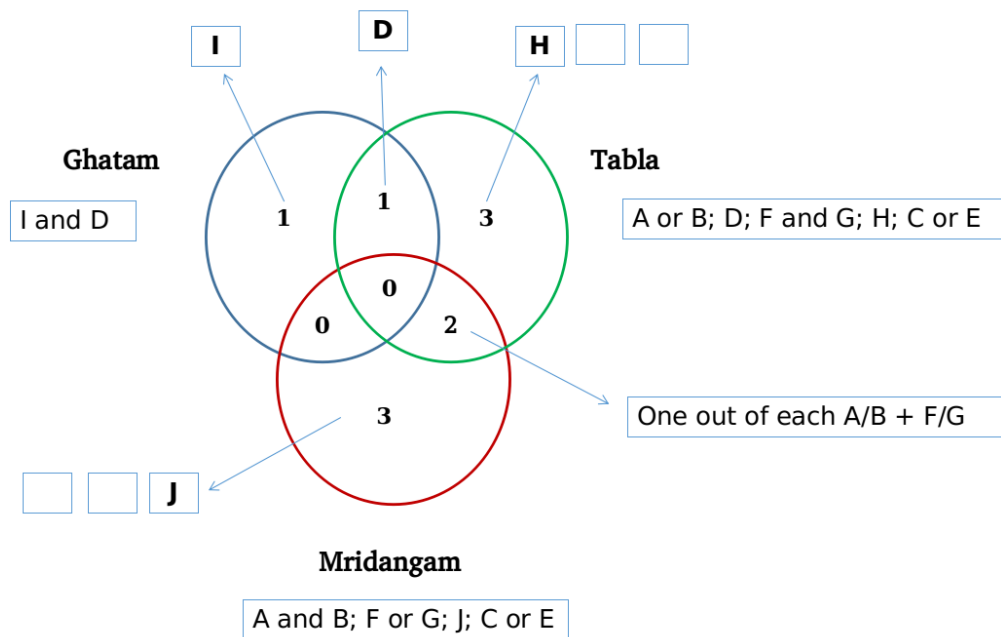
The conditions help us to further bifurcate the individuals based on their expertise.

Mridangam: A and B (condition 1); one out of F and G (condition 3)

Tabla: one out of A and B (condition 1); F and G (condition 3); D (condition 2)

Ghatam: D (condition 2)

Based on condition 4, we infer that 'I' and 'J' are either experts in Ghatam or Mridangam. However, condition 5 adds that 'I' is not an expert in Mridangam. This helps us definitively zero-in on 'I' as an expert in Ghatam. Since 'I' is a Ghatam expert, J is an expert in Mridangam and 'H' is an expert in Tabla [based on conditions 4 and 5]. Thus, we can depict our understanding so far as follows:



Mridangam [total: 5] - A and B (condition 1); one out of F and G (condition 3); J (condition 4 and 5); one out of C and E {remaining experts}

Tabla [total: 6] - one out of A and B (condition 1); F and G (condition 3); D (condition 2); H (condition 4 and 5); one out of C and E {remaining experts}

Ghatam [total: 2] - D (condition 2); I (condition 4 and 5)

Thus, we observe that **J** is definitely an expert in mridangam but not in either tabla or ghatam. Hence, Option B is the correct answer.

VIDEO SOLUTION

40.40

It is given that the total number of products supermarket sells is 320.

cosmetic + nutrition = foreign + domestic = FDA + EU = 320 products

In statement 1, it is given that the number of foreign products is equal to the number of domestic products.

Foreign products = Domestic products = $320/2 = 160$

Total(32)	Domestic (160)	Cosmetic	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
		Nutrition	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
	Foreign (160)	Cosmetic	EU -	
			FDA -	
		Nutrition	EU -	
			FDA -	

In statement 2, it is given that half of the domestic products were FDA approved cosmetic products, i.e. domestic, cosmetic and FDA = 80

In statement 4, it is given that there were 140 nutrition products, half of them were foreign products. This implies remaining half are domestic.

Total(32)	Domestic (160)	Cosmetic (90)	only EU -	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
	Foreign (160)	Cosmetic (90)	EU -	
			FDA -	
		Nutrition (70)	EU -	
			FDA -	

In statement 5, it is given that there are 200 FDA approved products out of which 70 are foreign products and 120 are cosmetic products.

If 70 are foreign products, remaining 130 should be domestic products. In domestic products, FDA approved cosmetic products are 80. This implies FDA approved nutrition products are $130-80$, i.e. 50.

There are 120 FDA approved cosmetic products.

Domestic, cosmetic and FDA approved = 80

This implies, Foreign, cosmetic and FDA approved is $120-80$, i.e. 40.

There are 70 FDA approved foreign products.

This implies Foreign, nutrition and FDA approved is $70-40$, i.e. 30.

Total(32)	Domestic (160)	Cosmetic (90)	only EU -	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU -	Total EU -
			only FDA -	Total FDA - 50
			Both -	
	Foreign (160)	Cosmetic (90)	EU - 50	
			FDA - 40	
		Nutrition (70)	EU - 40	
			FDA - 30	

Domestic and Cosmetic = 90

Domestic, cosmetic and FDA approved = 80

This implies, Domestic, cosmetic and FDA not approved is 90-80, i.e. 10.

Therefore, (domestic, cosmetic and only EU) = 10

Similarly, we get (domestic, nutrition and only EU) = 70-50 = 20

Total(32)	Domestic (160)	Cosmetic (90)	only EU - 10	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU - 20	Total EU -
			only FDA -	Total FDA - 50
			Both -	
	Foreign (160)	Cosmetic (90)	EU - 50	
			FDA - 40	
		Nutrition (70)	EU - 40	
			FDA - 30	

The number of foreign, cosmetic and FDA approved products is 40.

The answer is 40.

 VIDEO SOLUTION



CAT Syllabus PDF



41. D

It is given that the total number of products supermarket sells is 320.

cosmetic + nutrition = foreign + domestic = FDA + EU = 320 products

In statement 1, it is given that the number of foreign products is equal to the number of domestic products.

Foreign products = Domestic products = $320/2 = 160$

Total(32)	Domestic (160)	Cosmetic	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
		Nutrition	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
	Foreign (160)	Cosmetic	EU -	
			FDA -	
		Nutrition	EU -	
			FDA -	

In statement 2, it is given that half of the domestic products were FDA approved cosmetic products, i.e. domestic, cosmetic and FDA = 80

In statement 4, it is given that there were 140 nutrition products, half of them were foreign products. This implies remaining half are domestic.

Total(32)	Domestic (160)	Cosmetic (90)	only EU -	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
	Foreign (160)	Cosmetic (90)	EU -	
			FDA -	
		Nutrition (70)	EU -	
			FDA -	

In statement 5, it is given that there are 200 FDA approved products out of which 70 are foreign products and 120 are cosmetic products.

If 70 are foreign products, remaining 130 should be domestic products. In domestic products, FDA approved cosmetic products are 80. This implies FDA approved nutrition products are 130-80, i.e. 50.

There are 120 FDA approved cosmetic products.

Domestic, cosmetic and FDA approved = 80

This implies, Foreign, cosmetic and FDA approved is 120-80, i.e. 40.

There are 70 FDA approved foreign products.

This implies Foreign, nutrition and FDA approved is 70-40, i.e. 30.

Total(32)	Domestic (160)	Cosmetic (90)	only EU -	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU -	Total EU -
			only FDA -	Total FDA - 50
			Both -	
	Foreign (160)	Cosmetic (90)	EU - 50	
			FDA - 40	
		Nutrition (70)	EU - 40	
			FDA - 30	

Domestic and Cosmetic = 90

Domestic, cosmetic and FDA approved = 80

This implies, Domestic, cosmetic and FDA not approved is 90-80, i.e. 10.

Therefore, (domestic, cosmetic and only EU) = 10

Similarly, we get (domestic, nutrition and only EU) = $70 - 50 = 20$

Total(32)	Domestic (160)	Cosmetic (90)	only EU - 10	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU - 20	Total EU -
			only FDA -	Total FDA - 50
			Both -	
	Foreign (160)	Cosmetic (90)	EU - 50	
			FDA - 40	
		Nutrition (70)	EU - 40	
			FDA - 30	

The number of cosmetic products which do not have FDA approval = domestic only EU + foreign EU = $10 + 50 = 60$

The answer is option D.

[VIDEO SOLUTION](#)

42. C

It is given that the total number of products supermarket sells is 320.

cosmetic + nutrition = foreign + domestic = FDA + EU = 320 products

In statement 1, it is given that the number of foreign products is equal to the number of domestic products.

Foreign products = Domestic products = $320/2 = 160$

Total(32)	Domestic (160)	Cosmetic	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
		Nutrition	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
	Foreign (160)	Cosmetic	EU -	
			FDA -	
		Nutrition	EU -	
			FDA -	

In statement 2, it is given that half of the domestic products were FDA approved cosmetic products, i.e. domestic, cosmetic and FDA = 80

In statement 4, it is given that there were 140 nutrition products, half of them were foreign products. This implies remaining half are domestic.

Total(32)	Domestic (160)	Cosmetic (90)	only EU -	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
	Foreign (160)	Cosmetic (90)	EU -	
			FDA -	
		Nutrition (70)	EU -	
			FDA -	

In statement 5, it is given that there are 200 FDA approved products out of which 70 are foreign products and 120 are cosmetic products.

If 70 are foreign products, remaining 130 should be domestic products. In domestic products, FDA approved

cosmetic products are 80. This implies FDA approved nutrition products are 130-80, i.e. 50.

There are 120 FDA approved cosmetic products.

Domestic, cosmetic and FDA approved = 80

This implies, Foreign, cosmetic and FDA approved is 120-80, i.e. 40.

There are 70 FDA approved foreign products.

This implies Foreign, nutrition and FDA approved is 70-40, i.e. 30.

Total(32)	Domestic (160)	Cosmetic (90)	only EU -	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU -	Total EU -
			only FDA -	Total FDA - 50
			Both -	
	Foreign (160)	Cosmetic (90)	EU - 50	
			FDA - 40	
		Nutrition (70)	EU - 40	
			FDA - 30	

Domestic and Cosmetic = 90

Domestic, cosmetic and FDA approved = 80

This implies, Domestic, cosmetic and FDA not approved is 90-80, i.e. 10.

Therefore, (domestic, cosmetic and only EU) = 10

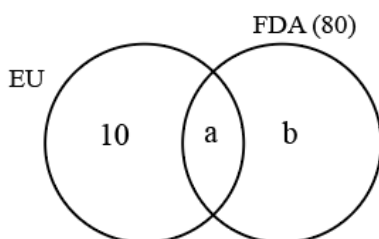
Similarly, we get (domestic, nutrition and only EU) = 70-50 = 20

Total(32)	Domestic (160)	Cosmetic (90)	only EU - 10	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU - 20	Total EU -
			only FDA -	Total FDA - 50
			Both -	
	Foreign (160)	Cosmetic (90)	EU - 50	
			FDA - 40	
		Nutrition (70)	EU - 40	
			FDA - 30	

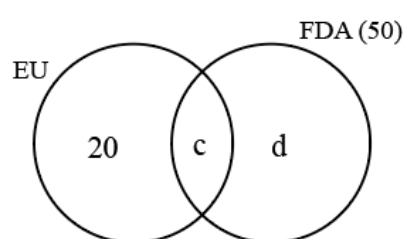
In the question, it is given that 70 cosmetic products did not have EU approval.

In foreign, 40 cosmetic products did not have EU approval. This implies 30 cosmetic products should have only FDA approval in domestic products.

Cosmetic (90)



Nutrition (70)

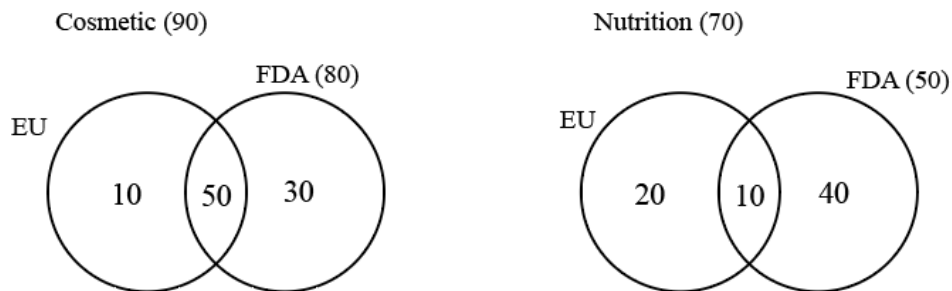


According to the above statement, $b = 30$

$a = 80 - 30 = 50$

Given, $a + c = 60$

$c = 60 - 50 = 10$



Therefore, the number of nutrition products which had both the approvals is 10.

The answer is option C.

[▶ VIDEO SOLUTION](#)

43.50

It is given that the total number of products supermarket sells is 320.

cosmetic + nutrition = foreign + domestic = FDA + EU = 320 products

In statement 1, it is given that the number of foreign products is equal to the number of domestic products.

Foreign products = Domestic products = $320/2 = 160$

Total(32)	Domestic (160)	Cosmetic	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
		Nutrition	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
	Foreign (160)	Cosmetic	EU -	
			FDA -	
		Nutrition	EU -	
			FDA -	

In statement 2, it is given that half of the domestic products were FDA approved cosmetic products, i.e. domestic, cosmetic and FDA = 80

In statement 4, it is given that there were 140 nutrition products, half of them were foreign products. This implies remaining half are domestic.

Total(32)	Domestic (160)	Cosmetic (90)	only EU -	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
	Foreign (160)	Cosmetic (90)	EU -	
			FDA -	
		Nutrition (70)	EU -	
			FDA -	

In statement 5, it is given that there are 200 FDA approved products out of which 70 are foreign products and 120 are cosmetic products.

If 70 are foreign products, remaining 130 should be domestic products. In domestic products, FDA approved cosmetic products are 80. This implies FDA approved nutrition products are $130-80$, i.e. 50.

There are 120 FDA approved cosmetic products.

Domestic, cosmetic and FDA approved = 80

This implies, Foreign, cosmetic and FDA approved is $120-80$, i.e. 40.

There are 70 FDA approved foreign products.

This implies Foreign, nutrition and FDA approved is 70-40, i.e. 30.

Total(32)	Domestic (160)	Cosmetic (90)	only EU -	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU -	Total EU -
			only FDA -	Total FDA - 50
			Both -	
	Foreign (160)	Cosmetic (90)	EU - 50	
			FDA - 40	
		Nutrition (70)	EU - 40	
			FDA - 30	

Domestic and Cosmetic = 90

Domestic, cosmetic and FDA approved = 80

This implies, Domestic, cosmetic and FDA not approved is 90-80, i.e. 10.

Therefore, (domestic, cosmetic and only EU) = 10

Similarly, we get (domestic, nutrition and only EU) = 70-50 = 20

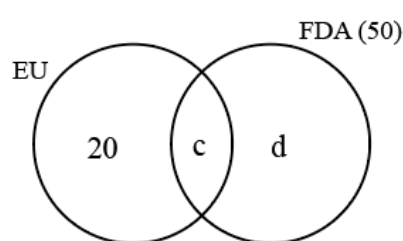
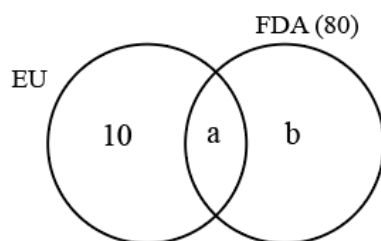
Total(32)	Domestic (160)	Cosmetic (90)	only EU - 10	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU - 20	Total EU -
			only FDA -	Total FDA - 50
			Both -	
	Foreign (160)	Cosmetic (90)	EU - 50	
			FDA - 40	
		Nutrition (70)	EU - 40	
			FDA - 30	

In the question, it is given that 50 nutrition products did not have EU approval.

In Foreign products, there are 30 nutrition products which do not have EU approval. This implies 20 nutrition products do not have EU(have only FDA) approval in domestic products.

Cosmetic (90)

Nutrition (70)



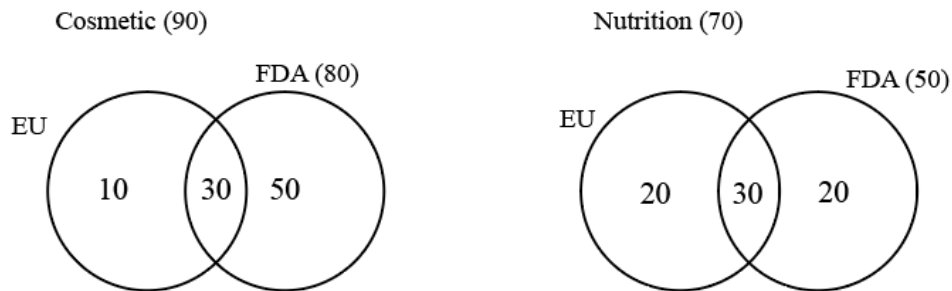
It is given, $d = 20$

$c = 50 - 20 = 30$

It is given, $a + c = 60$

$a = 60 - 30 = 30$

$b = 80 - 30 = 50$



Therefore, the number of domestic cosmetic products did not have EU(only FDA) approval is 50.

[VIDEO SOLUTION](#)

44. A

It is given that the total number of products supermarket sells is 320.

cosmetic + nutrition = foreign + domestic = FDA + EU = 320 products

In statement 1, it is given that the number of foreign products is equal to the number of domestic products.

Foreign products = Domestic products = $320/2 = 160$

Total(32)	Domestic (160)	Cosmetic	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
		Nutrition	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
	Foreign (160)	Cosmetic	EU -	
			FDA -	
		Nutrition	EU -	
			FDA -	

In statement 2, it is given that half of the domestic products were FDA approved cosmetic products, i.e. domestic, cosmetic and FDA = 80

In statement 4, it is given that there were 140 nutrition products, half of them were foreign products. This implies remaining half are domestic.

Total(32)	Domestic (160)	Cosmetic (90)	only EU -	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU -	Total EU -
			only FDA -	Total FDA -
			Both -	
	Foreign (160)	Cosmetic (90)	EU -	
			FDA -	
		Nutrition (70)	EU -	
			FDA -	

In statement 5, it is given that there are 200 FDA approved products out of which 70 are foreign products and 120 are cosmetic products.

If 70 are foreign products, remaining 130 should be domestic products. In domestic products, FDA approved cosmetic products are 80. This implies FDA approved nutrition products are $130-80$, i.e. 50.

There are 120 FDA approved cosmetic products.

Domestic, cosmetic and FDA approved = 80

This implies, Foreign, cosmetic and FDA approved is $120-80$, i.e. 40.

There are 70 FDA approved foreign products.

This implies Foreign, nutrition and FDA approved is 70-40, i.e. 30.

Total(32)	Domestic (160)	Cosmetic (90)	only EU -	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU -	Total EU -
			only FDA -	Total FDA - 50
			Both -	
	Foreign (160)	Cosmetic (90)	EU - 50	
			FDA - 40	
		Nutrition (70)	EU - 40	
			FDA - 30	

Domestic and Cosmetic = 90

Domestic, cosmetic and FDA approved = 80

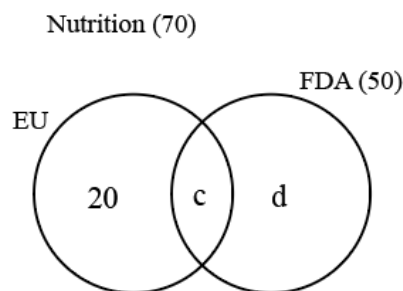
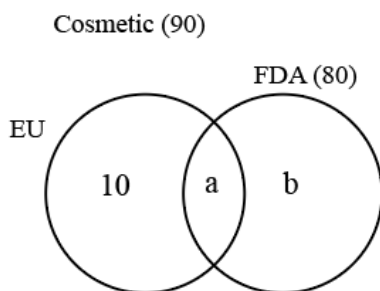
This implies, Domestic, cosmetic and FDA not approved is 90-80, i.e. 10.

Therefore, (domestic, cosmetic and only EU) = 10

Similarly, we get (domestic, nutrition and only EU) = 70-50 = 20

Total(32)	Domestic (160)	Cosmetic (90)	only EU - 10	Total EU -
			only FDA -	Total FDA - 80
			Both -	
		Nutrition (70)	only EU - 20	Total EU -
			only FDA -	Total FDA - 50
			Both -	
	Foreign (160)	Cosmetic (90)	EU - 50	
			FDA - 40	
		Nutrition (70)	EU - 40	
			FDA - 30	

In statement 3, it is given that the number of domestic products which have both the approvals = 60



In the question, it is given that $a + c = 60$

To find the minimum value of a , we need to maximise c .

Maximum value c can take is 50

Therefore, minimum value of a is 60-50, i.e. 10.

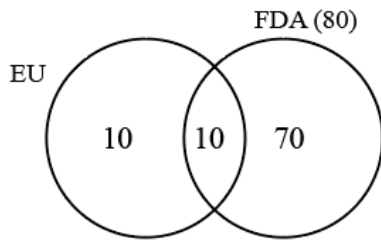
To find the maximum value of a , we need to minimise c .

Maximum value c can take is 0.

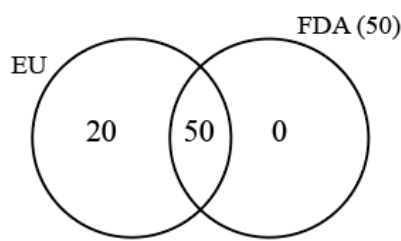
Therefore, maximum value of a is 60-0, i.e. 60.

a is minimum:

Cosmetic (90)



Nutrition (70)

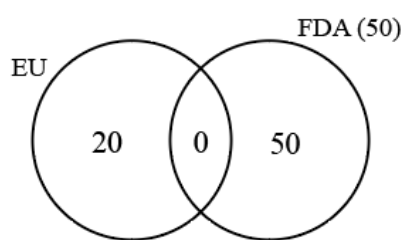


a is maximum:

Cosmetic (90)



Nutrition (70)



Therefore, the number of domestic cosmetic products that had both the approvals is at least 10 and at most 60.

The answer is option A.

[VIDEO SOLUTION](#)

45.20

Let the number of schools with exactly three of the facilities was the same irrespective of which three were considered be x .

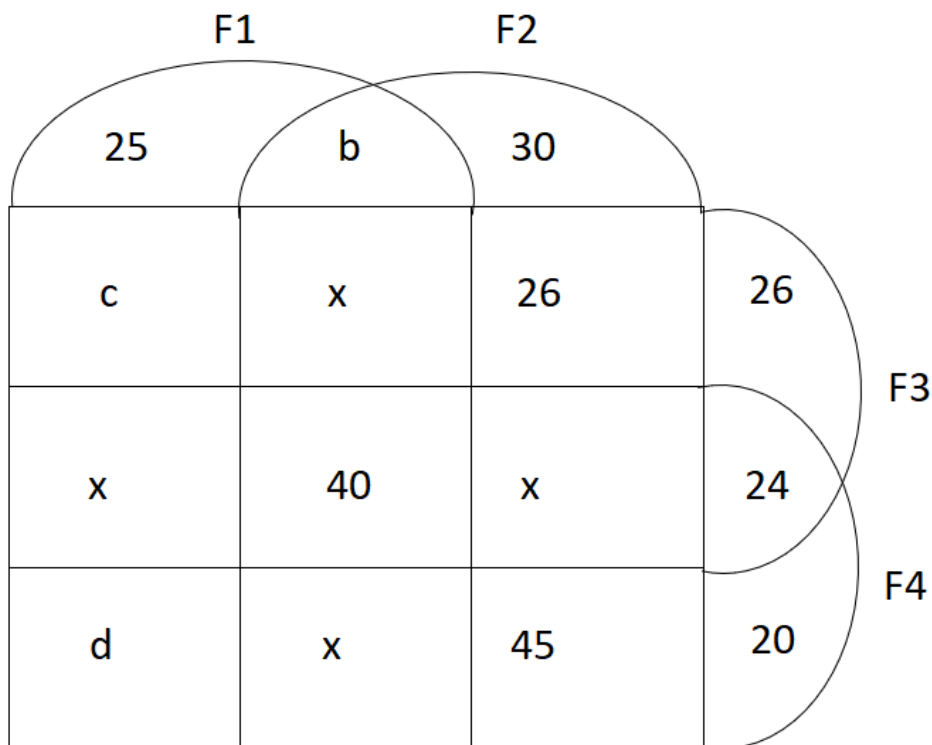
Number of schools with none of the facilities be 'n' from 1, $n=80$.

Number of schools with only F1 and F2 be 'b'

Number of schools with only F1 and F3 be 'c'

Number of schools with only F1 and F4 be 'd'

From the information given in the question we will get the following Venn diagram.



From 5, $b+141+3x=313 \Rightarrow b+3x=172 \dots (i)$

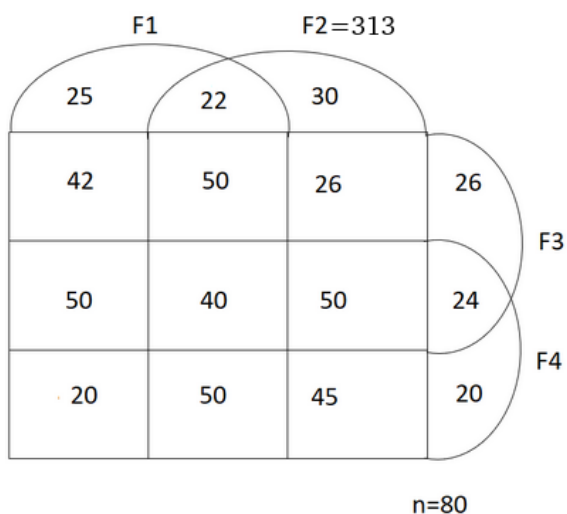
From 8, $b+x+40+x=162 \Rightarrow b+2x=122 \dots (ii)$

(ii)-(i) gives $x=50 \Rightarrow b=22$

From 9, $237+3x+c+d=279+3x=d \Rightarrow c=42$

Total number of schools = 600 $\Rightarrow 313+25+c+x+d+26+24+20+80=600 \Rightarrow d=20$.

The final table looks like:



The total number of schools having only F1 and F4=20.

[VIDEO SOLUTION](#)



46. A

Let the number of schools with exactly three of the facilities was the same irrespective of which three were considered be x .

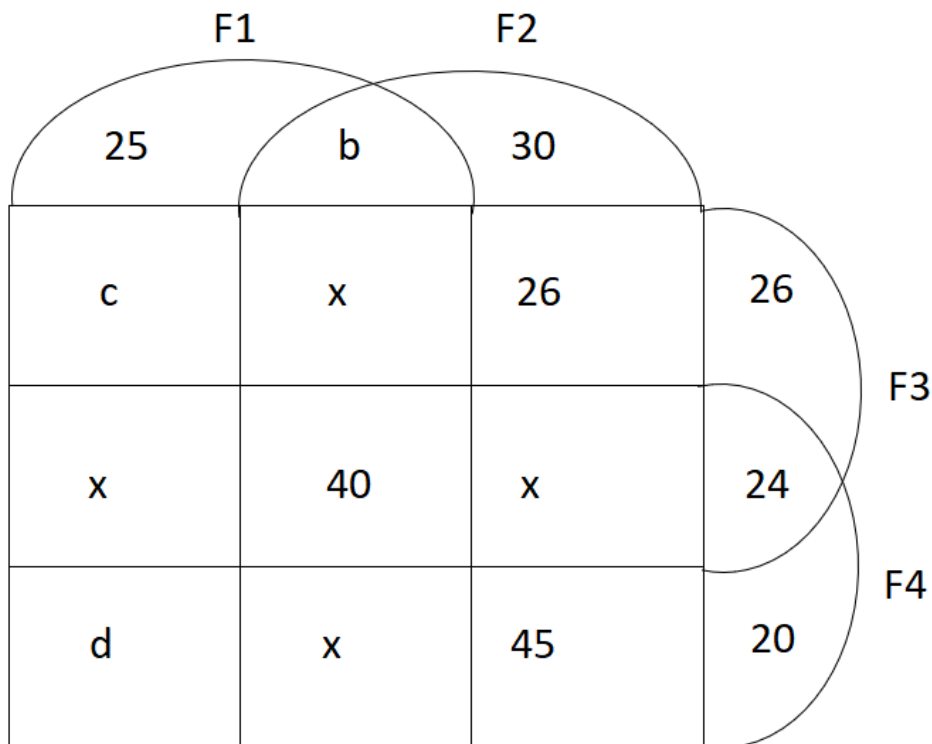
Number of schools with none of the facilities be 'n' from 1, $n=80$.

Number of schools with only F1 and F2 be 'b'

Number of schools with only F1 and F3 be 'c'

Number of schools with only F1 and F4 be 'd'

From the information given in the question we will get the following Venn diagram.



From 5, $b+141+3x=313 \Rightarrow b+3x=172 \dots (i)$

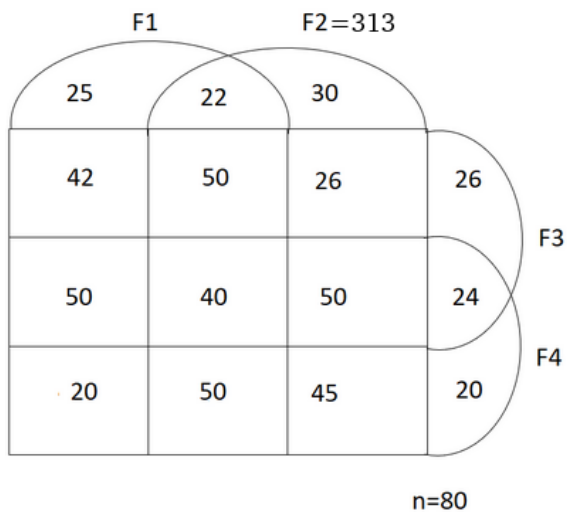
From 8, $b+x+40+x=162 \Rightarrow b+2x=122 \dots (ii)$

(ii)-(i) gives $x=50 \Rightarrow b=22$

From 9, $237+3x+c+d=279+3x=d \Rightarrow c=42$

Total number of schools = 600 $\Rightarrow 313+25+c+x+d+26+24+20+80=600 \Rightarrow d=20$.

The final table looks like:



The total number of schools having facilities F4 and F2= $45+50+50+40=185$.

[VIDEO SOLUTION](#)

47. C

Let the number of schools with exactly three of the facilities was the same irrespective of which three were considered be x .

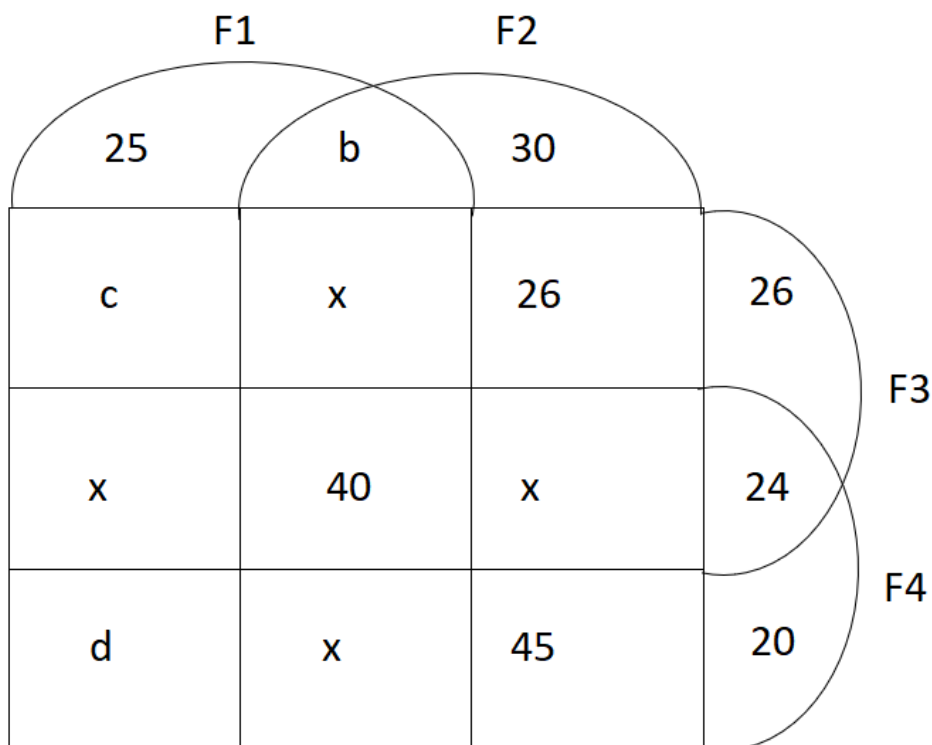
Number of schools with none of the facilities be 'n' from 1, $n=80$.

Number of schools with only F1 and F2 be 'b'

Number of schools with only F1 and F3 be 'c'

Number of schools with only F1 and F4 be 'd'

From the information given in the question we will get the following Venn diagram.



From 5, $b+141+3x=313 \Rightarrow b+3x=172 \dots (i)$

From 8, $b+x+40+x=162 \Rightarrow b+2x=122 \dots (ii)$

(ii)-(i) gives $x=50 \Rightarrow b=22$

From 9, $237+3x+c+d=279+3x=d \Rightarrow c=42$

Total number of schools = 600 $\Rightarrow 313+25+c+x+d+26+24+20+80=600 \Rightarrow d=20$.

The final table looks like:

F1		F2=313		
25	22	30		
42	50	26	26	
50	40	50	24	F3
20	50	45	20	F4
n=80				

The total number of schools with exactly three of the four facilities = $4x=200$.

[VIDEO SOLUTION](#)

48.42

Let the number of schools with exactly three of the facilities was the same irrespective of which three were considered be x .

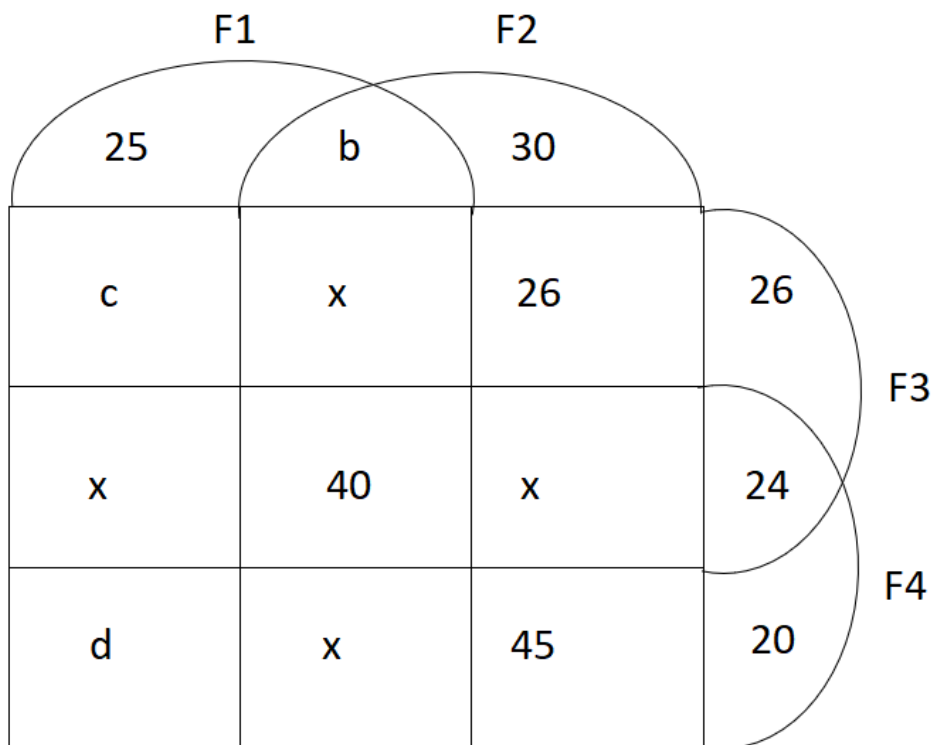
Number of schools with none of the facilities be ' n ' from 1, $n=80$.

Number of schools with only F1 and F2 be ' b '

Number of schools with only F1 and F3 be ' c '

Number of schools with only F1 and F4 be ' d '

From the information given in the question we will get the following Venn diagram.



From 5, $b+141+3x=313 \Rightarrow b+3x=172 \dots (i)$

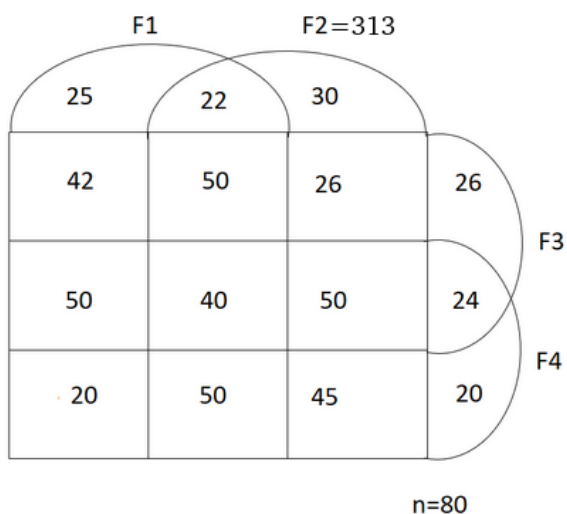
From 8, $b+x+40+x=162 \Rightarrow b+2x=122 \dots (ii)$

(ii)-(i) gives $x=50 \Rightarrow b=22$

From 9, $237+3x+c+d=279+3x=d \Rightarrow c=42$

Total number of schools = 600 $\Rightarrow 313+25+c+x+d+26+24+20+80=600 \Rightarrow d=20$.

The final table looks like:



The total number of schools having only F1 and F3 = $c=42$

[VIDEO SOLUTION](#)